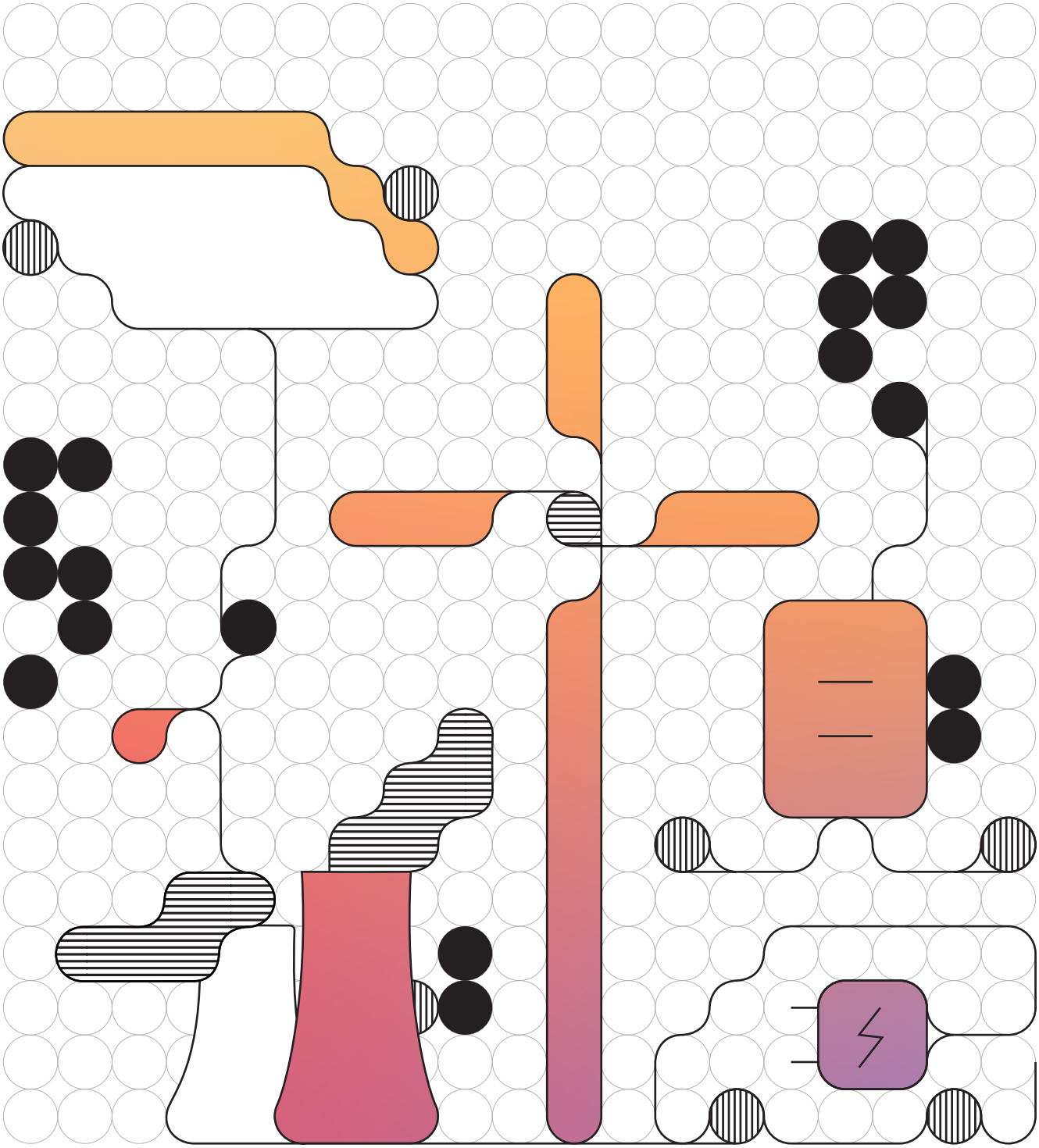
Digital Energy Grid

A vision for a unified energy infrastructure



FIDE
Foundation for Interoperability in Digital Economy

With contributions from
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Digital Energy Grid
A vision for a unified energy infrastructure

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About FIDE & IEA

FIDE

FIDE, or Foundation for Interoperability in Digital Economy (fide.org), is a not-for-profit organisation that fosters an open, inclusive, decentralised and fair digital economy through interoperable open standards and architecture. FIDE achieves this by promoting innovation and co-creation among participants in the digital ecosystem through the interoperable, open specifications for digital economic transactions, **Beckn Protocol** (becknprotocol.io) (or Beckn in short), of which it is the genesis author.

Beckn Protocol, first published in October 2019, is governed under a Creative Commons Licence and maintained as a digital public good. FIDE promotes Beckn globally in a pro-bono advisory role to enable innovation and co-creation among ecosystem participants. Beckn represents a paradigm shift from platforms to open protocols, offering universality and decentralisation modelled on the design principles of the Internet.

IEA

IEA, or the International Energy Agency (iea.org), works with governments and industry to shape a secure and sustainable energy future for all. The IEA is at the heart of global dialogue on energy, providing authoritative analysis, data, policy recommendations, and real-world solutions to help countries provide secure and sustainable energy for all. The IEA was created in 1974 to help co-ordinate a collective response to major disruptions in the supply of oil. While oil security remains a key aspect of our work, the IEA has evolved and expanded significantly since its foundation.

Taking an all-fuels, all-technology approach, the IEA recommends policies that enhance the reliability, affordability and sustainability of energy. It examines the full spectrum issues including renewables, oil, gas and coal supply and demand, energy efficiency, clean energy technologies, electricity systems and markets, access to energy, demand-side management, and much more.

Since 2015, the IEA has opened its doors to major emerging countries to expand its global impact, and deepen cooperation in energy security, data and statistics, energy policy analysis, energy efficiency, and the growing use of clean energy technologies.

Foreword



Nandan Nilekani
Co-founder
FIDE

We stand at a pivotal moment in the global energy transition. The convergence of decentralisation, digitalisation, and surging demand is reshaping the energy landscape, presenting unprecedented opportunities alongside formidable challenges. At the heart of this transformation lies the need to seamlessly manage three critical flows: information, energy, and transactions across a decentralised and diverse ecosystem. Addressing these challenges requires more than just advanced physical infrastructure—it demands a unified, interoperable framework that integrates data, resources, and stakeholders. This is where the vision of the Digital Energy Grid (DEG) becomes truly transformative.

DEG envisions a future where universal digital building blocks—identity, machine-readable data, and verifiable transactions—create an interoperable backbone for the energy ecosystem. By digitising assets, participants, and transactions, DEG addresses systemic inefficiencies, reduces transactional frictions, and unlocks cascading benefits across industries. This approach moves beyond isolated solutions, embracing a shared, scalable framework that can trigger a flywheel effect of innovation and combinatorial value.

One of the most exciting aspects of the DEG is its potential to enable symbiotic relationships between energy and technology. Take, for instance, the synergy between DEG and Artificial Intelligence (AI). By harmonising fragmented data streams and enabling transaction-ready datasets, DEG enhances AI's contextual intelligence, driving superior strategic insights and real-time decision-making. In turn, AI amplifies DEG's potential, enabling adaptive energy management, precise demand forecasting, and dynamic grid optimisation. Together, they can establish a dynamic, agentic energy ecosystem where AI-driven systems autonomously optimise supply-demand equilibria, enable real-time market transactions, and seamlessly integrate information, energy, and financial flows at scale.

This paper marks a significant step toward realising this vision in collaboration with the International Energy Agency (IEA), an organisation at the forefront of global energy cooperation. The IEA's leadership in fostering international collaboration and bringing together diverse perspectives is pivotal to navigating the complexities of the energy transition. Their involvement underscores the importance of a coordinated, global approach to building the energy systems of the future.

However, realising this vision will require collective action. I invite stakeholders across the energy ecosystem—governments, industries, innovators, and communities—to join us in shaping this transformative future. Together, we can build a Digital Energy Grid that not only addresses the challenges of today but also lays the foundation for a sustainable, resilient, and inclusive energy future. The time to act is now!



Dr. Fatih Birol
Executive Director
IEA

Digitalisation is a key trend across our economies and the energy sector is no exception: vast quantities of data are gathered, interpreted and acted upon every day to manage the increasing complexity of energy markets.

In our 2024 World Energy Outlook we showed that the world is entering into the Age of Electricity, as this unique "fuel" plays an increasingly crucial role in all economies in the world. Demand for electricity is set to grow six times faster than overall energy demand to 2035. This is due partly to traditional uses like air conditioning and industrial motors, but also to new sectors such as electric vehicles and data centres.

One aspect the Age of Electricity is the growth of cost competitive renewable sources of electricity generation. In 2024, 80% of global capacity additions of electricity generation capacity came from low-emissions sources, and solar PV is leading the way. In 2024, the world added so much solar PV capacity that the output from new installations in this year alone could power the entire country of Korea. At the same time, generation from the world's nuclear reactors is on track to reach new heights in 2025.

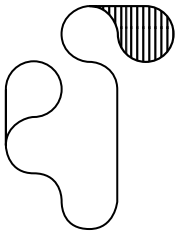
These trends create huge opportunities to deliver the energy people need, more efficiently, affordably and productively, but also with lower emissions. But it also creates challenges.

One of these is balancing an increasingly variable, decentralised and digitalised electricity grid. IEA analysis shows that the demand for electricity system flexibility, or the ability of the electricity system to adjust supply and demand to ensure they always match, will increase by 2.5 times in Europe to 2035. Meeting this will require flexibility from all parts of the system, with a growing contribution from the smart combination of battery storage and demand response.

This will require facilitating the ability of system operators to communicate with millions, indeed billions, of distributed devices, from behind-the-meter battery storage to electric vehicles to air conditioning units. It will necessitate flows of data, new markets, and new incentives.

At the IEA, we have always welcomed big new ideas. We need them to solve the challenges we face and realise the opportunities ahead. FIDE's idea of a "Digital Energy Grid" – an overarching digital framework that facilitates more interconnectivity between energy assets – is an example. FIDE's work – including this new report, to which the IEA is pleased to have contributed – provides thought leadership about how we can design the long-term frameworks to rise to these changes. Bringing these perspectives together is key to find solutions.

Executive summary



The global energy landscape is undergoing a transformative shift, driven by decentralisation, digitalisation, and surging demand from electrification and economic growth. These shifts are reshaping how energy is produced, distributed, and consumed, presenting both unprecedented opportunities and complex challenges.

By 2035, nearly 3668 GW of rooftop solar PV is expected to be installed globally—a staggering 429% increase from today[1]. This expansion underscores the promise of decentralized renewables but also highlights the difficulty of managing distributed and variable resources. Global electricity demand is projected to nearly double by 2050[2] under current policies and could rise even further under more ambitious scenarios. Key contributors include the electrification of heating, transport, industrial processes and increased energy demands from data centres. Emerging markets and developing economies are expected to drive nearly 80% of this growth by 2030[3].

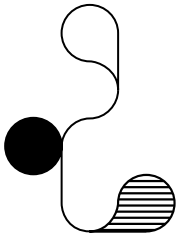
This rapid demand growth and the proliferation of distributed energy resources (DERs) expose the urgent need for better coordination across the energy ecosystem. Technologies such as heat pumps and air conditioners are increasing demand sensitivity to temperature fluctuations, while extreme weather events like heatwaves place additional strain on power systems. Electrification is also altering hourly demand patterns, with peak demand rising faster than average demand in many locations —requiring significant investment in grid infrastructure.

While policy-driven, market-based, and technological solutions exist to address these challenges, many are **purpose-specific**, targeting isolated issues rather than systemic needs. Current solutions, often fragmented and non-interoperable impede the seamless integration of resources and stakeholders. For instance, limited grid visibility complicates the integration of variable renewables and DERs. Furthermore, the **reliance on non-digital modes of operation** exacerbates these challenges. Many processes lack portable, verifiable digital records, leading to information asymmetry and trust issues. Physical certifications for equipment that cannot be digitally verified and energy transactions without digital records hinder transparency, increase costs, and limit coordination. These fragmented and non-digital practices leave the ecosystem ill-equipped to adapt to the complexities of the energy transition.

Digital solutions hold immense potential to facilitate a smooth energy transition, yet their impact is diminished when developed in silos. **The energy ecosystem faces three interconnected coordination challenges of managing information flows, energy flows, and transaction flows seamlessly - all in a more decentralised energy systems of the future.**

Addressing these challenges requires more than advanced physical infrastructure—it demands robust frameworks for sharing critical data such as demand, supply, and pricing signals. Advancements in smart devices, data analytics, and AI-based forecasting offer new opportunities, but physical energy systems remain constrained by technical, financial, and infrastructural limitations.

The true challenge lies in synchronising the flow of information with the physical flow of energy across diverse stakeholders and distributed resources. **A transformative solution lies in adopting an infrastructure-led approach: establishing a foundational digital infrastructure that moves beyond piecemeal solutions to embrace a shared, scalable framework.** This concept can be envisioned as a data-grid to augment the physical energy grid, designed to handle the complexity and variability of modern energy markets.



At the heart of this approach is the vision for the Digital Energy Grid (DEG) that unifies the energy ecosystem through universal digital building blocks. DEG introduces key principles such as universal identity, machine readability, and verifiability, creating a composable, programmable, and portable digital infrastructure. By digitising transactions, assets, and participants in an interoperable manner across the value chain, DEG creates a digital backbone of the energy ecosystem that enhances transparency, reliability, and adaptability.

For example, DEG assigns universal identities to energy resources and actors, akin to phone numbers or postal codes, ensuring global uniqueness. It enables machine readability by digitising essential metadata such as energy ratings and certifications, enhancing efficiency and interoperability. Through cryptographic verifiability, DEG ensures tamper-proof digital records, fostering trust while preserving data privacy and sovereignty.

These capabilities unlock systemic benefits: composability, programmability, and portability of assets and data. DEG integrates core functionalities like observability and secure transactions into the infrastructure, aligning the physical flow of energy with digital information and transaction flows. This approach facilitates innovation and collaboration at scale, streamlining progress toward shared energy goals.

DEG is not a product or standalone technology but a vision for an interoperable, scalable, and modular digital infrastructure. It complements existing systems, bridging legacy standards while enabling future advancements. By reducing fragmentation and fostering a cohesive framework, DEG creates conditions for private innovation and public policy to co-evolve.

Much like a road network enables trade across regions, DEG envisions a digital network that seamlessly coordinates energy, data, and value. This unified approach potentially addresses systemic inefficiencies, reduces transaction frictions, and unlocks cascading benefits essential for a sustainable energy transition.

The Digital Energy Grid offers a critical pathway to energy system transformation—unifying purpose-specific innovations into a cohesive energy future capable of addressing both anticipated and unforeseen challenges of the transition.

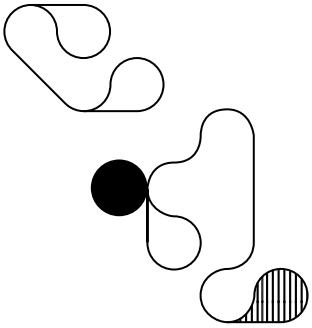


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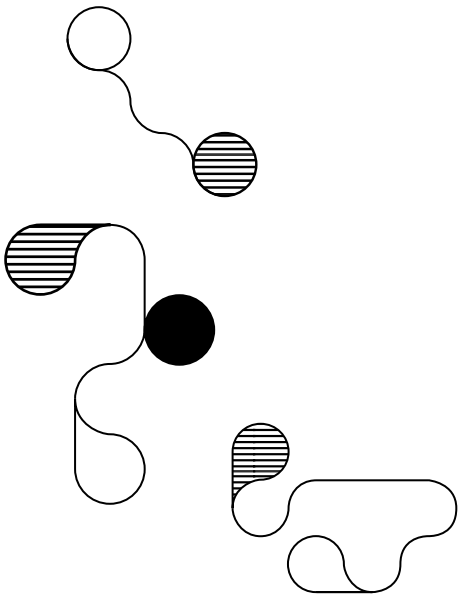
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Introduction

The energy landscape is undergoing a profound transformation, driven by the convergence of decentralization, digitalization, and rising demand fueled by electrification and economic growth. This shift presents vast opportunities alongside critical challenges, including managing distributed and variable resources, rising peak demand, and the integration of climate-sensitive technologies like heat pumps. Addressing these complexities demands a coordinated, infrastructure-led approach centered on a scalable digital energy grid.

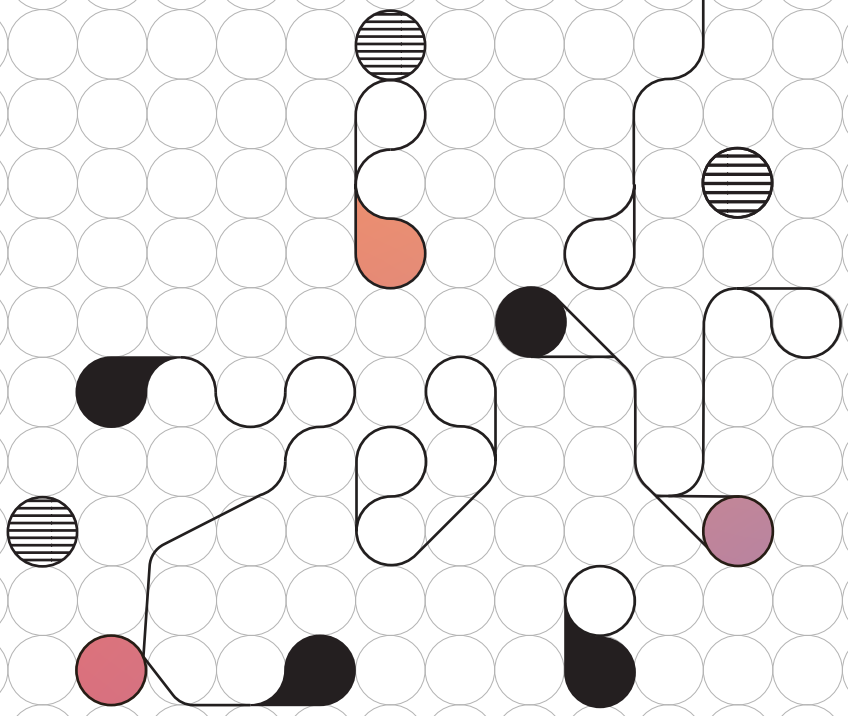
This paper presents a vision for a unified energy ecosystem capable of addressing the structural barriers and coordination gaps that currently impede progress in the energy sector. It first explores why the current energy system remains fragmented and fails to scale to the levels necessary for a sustainable future. It examines the limitations of existing solutions and the systemic barriers to progress. Next, the paper introduces an infrastructure-led approach as a compelling alternative to isolated solutions. This paradigm shift toward infrastructural thinking underpins the foundation of the Digital Energy Grid (DEG). The paper explores the specifics of DEG—what it is, what it is not, and the transformative possibilities it unlocks for the energy ecosystem of tomorrow. Finally, it outlines how DEG can be implemented at a high level.

The paper presents a vision, not a definitive solution, aiming to spur global dialogue and collaboration among stakeholders to advance this approach, amplify ongoing efforts, and progress toward a unified architecture for a more efficient, sustainable, and equitable energy ecosystem.



01

We are in the middle of one of the **biggest energy transformations in history**



The global energy landscape is undergoing one of the most profound transformations in history. This is driven by an urgent need for decarbonization, rapid growth in energy demand, and breakthroughs in technology that are changing how electricity is produced, distributed, and consumed.

An estimated 748.5 million people worldwide still lack access to electricity[4], a reminder of the challenges facing today's energy sector: ensuring universal access while pursuing a secure, resilient, and sustainable energy future. Addressing these challenges will require a comprehensive approach—significant investments in physical grid infrastructure to extend access to underserved regions, along with the deployment of clean energy systems such as renewables to expand generation capacity in a sustainable way. A future-ready energy infrastructure becomes essential to support this increasingly decentralized landscape.

1.1
Energy demand is soaring with electrification, economic growth, heating and cooling needs. Meeting this growing demand sustainably is a challenge that highlights the need for greater efficiency

Recent data show that over the past decade, global energy demand increased at an average rate of 1.4% per year, with emerging economies hitting rates as high as 2.56%. Global electricity demand growth in 2024 was forecast to reach 4.2%, among the highest levels in the past two decades. Under current policies, electricity demand could nearly double by 2050, with more ambitious policy scenarios likely driving it higher still[5]. Various forces contribute to this increase, including the electrification of heating, transport, and industrial processes, rising demand from data centers, and a higher prevalence of household appliances. Although renewables continue to grow their share of the electricity mix, the pace must accelerate substantially to meet international climate targets and still ensure reliable, affordable, and secure electricity access—especially in low- and middle-income countries.

Electrification of vehicles

Electric mobility is a major driver reshaping electricity usage. The market share of electric vehicles (battery electric and plug-in hybrids) was expected to be around 20.9% of global sales in 2024 and could climb to 43.4% by 2030. By then, nearly one in six cars in the global fleet is projected to be electric, with a

combined potential energy storage capacity of over 16 TWh— equivalent to over six times the global BESS (Battery Energy Storage System) capacity of 2.46 TWh in 2030[6]. Although EVs raise power demand, they also offer flexibility: with vehicle-to-grid (V2G) technologies, individual owners can supply stored battery energy back to the grid during peak times if robust energy management systems are in place.

Increasing heating and cooling needs

The year 2024 marked the hottest on record—the first time the world crossed the 1.5-degree threshold, with the global average temperature hitting 1.6 degrees above pre-industrial levels[7]. As incomes rise and heatwaves and extreme weather become more frequent, electricity demand for cooling is set to surge, turning air-conditioning from a luxury into a necessity for millions. Residential demand for space cooling could more than triple by 2050 under current policy scenarios. By 2030, the global air-conditioner stock is likely to exceed 2.1 billion units, a 36.8% climb from today. Heat pump adoption is also on the rise, with their share in residential heating equipment sales expected to reach 22% by 2030, for a total capacity of 1.88 TW[8]. All this underscores how weather-driven loads are increasingly shaping electricity demand patterns.

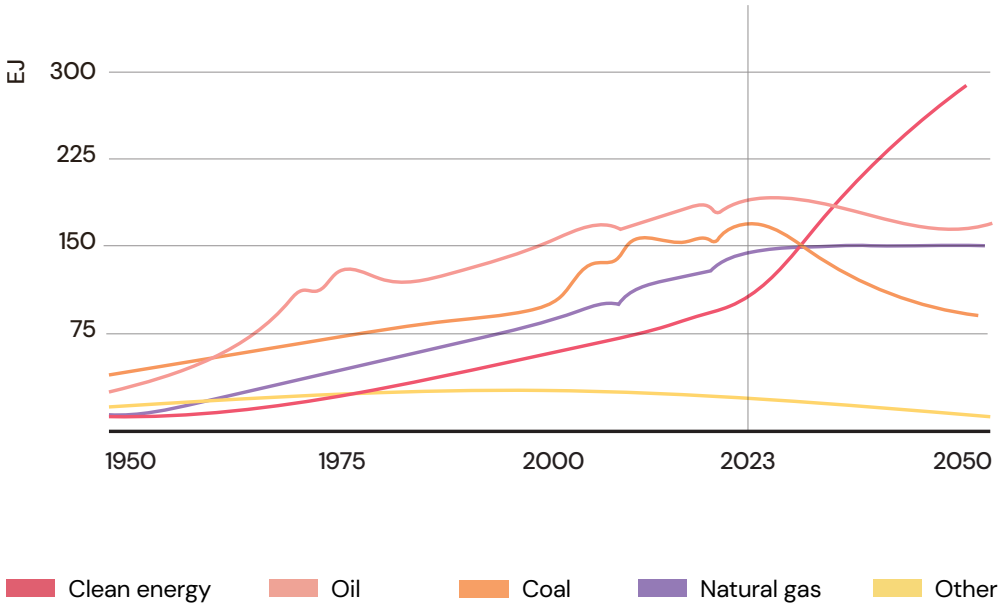
Data centers

The energy demands of data centers are evolving alongside advancements in AI. While training AI models has historically driven exponential growth in computational power—doubling roughly every five to six months since 2010[9] —recent innovations, such as more comprehensive architectures and new inference techniques, suggest that this trend may not be linear. Nevertheless, large-scale data centers continue to consume vast amounts of electricity, sometimes equivalent to that of 100,000 households[10]. While global data centers account for around 1% of total electricity consumption[11], the load can be heavily concentrated regionally. In Virginia, for instance, data centers already account for over a quarter of the state's total electricity consumption[12]. Technology companies often aim to power these facilities with renewables, making them some of the largest corporate purchasers of renewable energy—but that requires accurate, timely information on grid capacity and permitting rules. As data centers expand rapidly, planners face the possibility that renewable capacity may not keep pace with new demand, raising the risk of local or even regional energy shortages. This dynamic brings new urgency to the need for coordination across infrastructure, finance, and policy to secure adequate, sustainable electricity supplies.

1.2
Innovations in variable renewables, distributed resources, and batteries are reshaping energy systems—but also testing grid stability

Growth in variable renewables

One of the most important developments is the increase in variable sources of renewable electricity generation, especially from solar PV and wind. Generation from solar PV and wind combined is projected to nearly triple between 2023 and 2030 under current policy trends, with solar PV alone driving roughly 80% of that capacity growth. In 2023, fossil fuels constituted 60.1% of global electricity supply, typically from centralized thermal power plants, but that share is projected to decline to 44% by 2030. Over the same period, capacity additions in solar PV and wind are forecast to contribute over 90% of the net increase in electricity supply, pointing to a marked shift in the global power mix. By 2030, wind and solar together are expected to comprise about 30.6% of global electricity supply and substantially higher proportions in certain key markets—44.1% in Europe, 24.4% in India, 38.5% in China, and 34.5% in the United States[13].



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Figure 1. Global energy mix by scenario to 2050 in the Stated Policies Scenario - A scenario based on today's policy settings sees clean energy poised for huge growth, while coal, oil and natural gas each reach a peak by 2030 and then start to decline[14].

Notes. EJ - exajoules. Oil, coal and natural gas refer to unabated uses as well as non-energy use. Clean energy includes renewables, modern bioenergy, nuclear, abated fossil fuels, low-emissions hydrogen and hydrogen-based fuels. Other includes traditional use of biomass and non-renewable waste.

Increasing use of distributed technologies

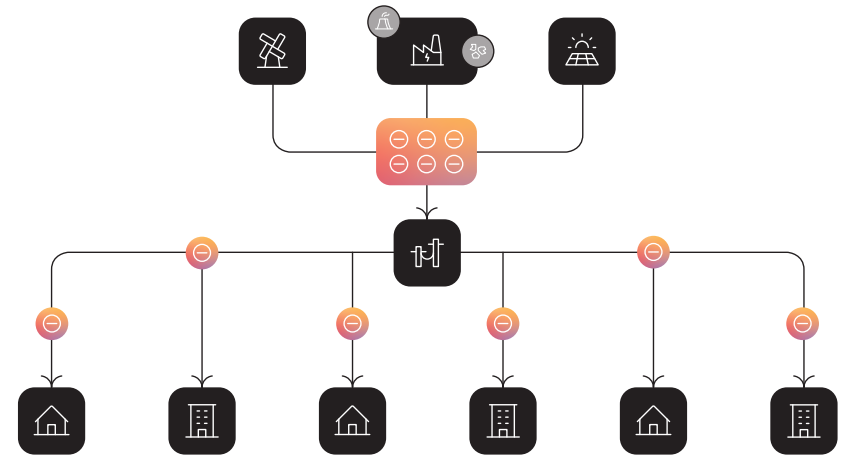
Within the trend of variable renewables, distributed energy technologies like rooftop solar PV and community wind projects are growing in importance. As of today, installed global solar PV capacity stands at about 1,610 GW, with roughly 43.1% as rooftop solar. By 2035, total rooftop and utility-scale solar PV installations could increase by over five times, pushing rooftop solar specifically to around 3,700 GW—an enormous 429% jump from current levels[15]. Distributed energy resources (DERs) bring generation closer to load centers, reducing transmission losses while enhancing grid resilience. Yet managing widespread, small-scale, and intermittent sources poses operational and regulatory challenges across many parts of the world.

Energy storage is proving pivotal for integrating higher shares of renewables, helping to smooth out intermittency. While pumped hydro and other forms of energy storage remain significant, battery energy storage systems (BESS) are expected to account for 90% of the growth in storage capacity in the coming years[16]. Battery storage capacity is projected to increase nearly tenfold from 88.8 GW in 2023 to over 850 GW, with the average duration increasing from around two hours today to nearly three hours in 2030. Approximately 38% of current BESS capacity—about 33 GW—operates behind the meter (for example, in homes and businesses). By 2030, the total behind-the-metre BESS power capacity is expected to grow more than fivefold to nearly 180 GW, with over half of this capacity in China. Utility-scale batteries could increase by over twelvefold, from 55.1 GW now to more than 675.3 GW by 2030[17]. These trends reinforce that DERs and battery innovations can bring major benefits in resilience and adaptability for a low-carbon transition.

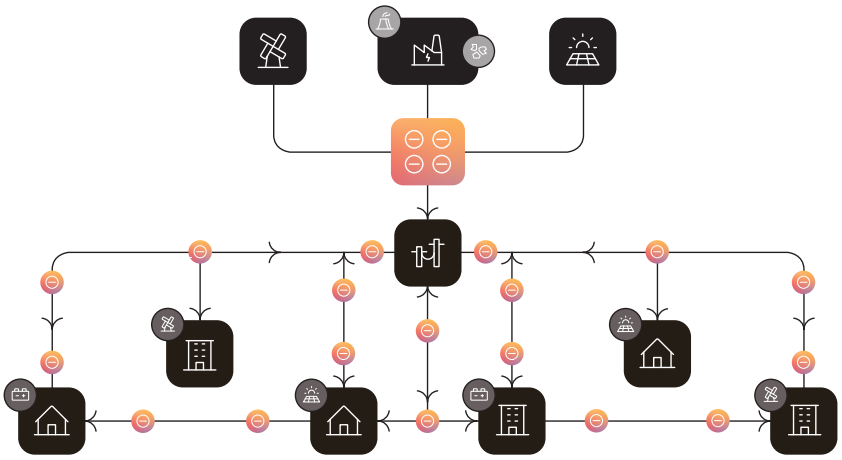
Demand response and grid management enabled by digitalization

Digital technologies—smart meters, demand-response-enabling appliances, and IoT-based controls—are multiplying, making it possible to engage end users in grid operations in ways previously unthinkable. Tools such as Grid Orchestration Software, Network Modelling, and Distributed Energy Resource Management Systems (DERMS) enable real-time monitoring and grid flexibility, supporting renewable integration especially in high-income countries. By 2022, the number of smart power meters worldwide surpassed one billion, a tenfold jump from 2010. Meanwhile, connected devices with automated controls and sensors are expected to reach 13 billion in 2023, up from fewer than 1 billion a decade ago[18]. Early examples include peer-to-peer energy trading pilots in Australia[19] and solar buy-back schemes in parts of the United States[20], which allow consumers to transact at micro scales. As these approaches expand, dynamic pricing is likely to become more prevalent, potentially boosting the responsiveness of end users.

1.3
There is a trend towards higher volume, more micro-value, dynamic and bi-directional transactions in the energy market, underscoring the need to modernise the grid



i. Static, mega, uni-directional



ii. Dynamic, micro, bi-directional

Figure 2. Micro, bidirectional transactions are redefining the future of inclusive, high-volume energy exchanges.

Amid the growth of DERs, energy transactions are becoming more frequent, smaller in value, increasingly time-sensitive, and bidirectional. Where consumption used to be largely passive and predictable, consumers in some regions now actively participate in flexible demand programs or generate surplus power with rooftop solar. Traditional grid setups are not yet fully prepared for this shift, reinforcing the need for modern, interoperable systems.

1.4
Infrastructure strain – Blackouts, negative pricing, and interconnection bottlenecks

Mounting pressure on today's grids manifest in several ways –

Power outages

Regions from Latin America to South Asia, Europe, and North America have experienced severe blackouts of increasing frequency and duration, revealing vulnerabilities in adapting to extreme weather and crises. Unreliable forecasting and inflexible response can drive unexpected price spikes and widespread disruptions. These outages cost around $100 billion each year—about 0.1% of global GDP[21].

Negative energy prices and curtailment

Simultaneously, the rise of renewables has introduced a new challenge: negative energy prices. In 2024, the share of negative priced hours in Southern California exceeded 20% marking a more than threefold increase compared to the previous year[22]. This issue is compounded when renewable producers are compensated to curtail output – a stopgap that is financially and environmentally inefficient compared to encouraging flexible-demand response and storage solutions.

Grid congestion

Delays in connecting new renewable projects and updating grid infrastructure are emerging as critical barriers. In Germany, congestion management costs exceeded EUR 4 billion in 2022—funds that are equivalent to investing in about 4.5 gigawatts (GW) of new solar PV capacity. In the United States, transmission grid congestion costs ballooned from USD 6 billion in 2019 to nearly USD 21 billion in 2022. Britain spent nearly GBP 1 billion just for balancing actions in winter 2021–2022 in response to transmission constraints[23]. Such challenges, if left unresolved, could hamper both energy security and climate aspirations.

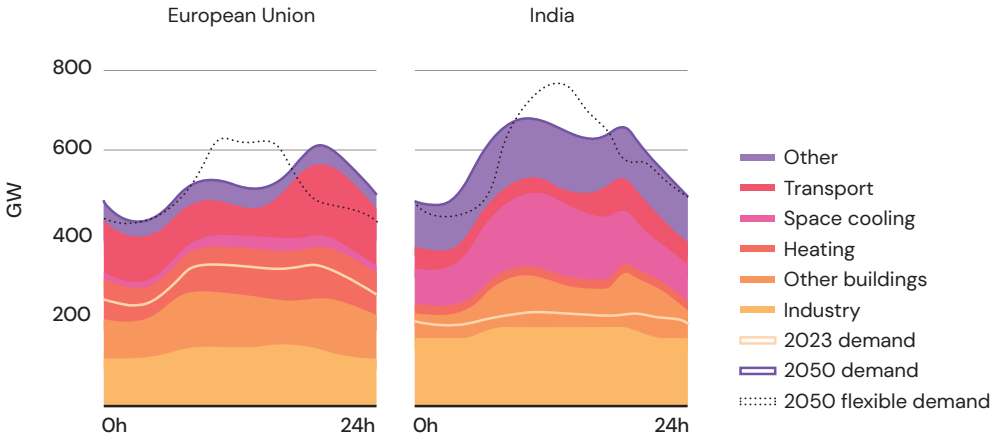
Rising incidents and growing financial impacts underline the need for more flexible supply and demand capabilities. However, a mix of technical, economic, and informational constraints often blocks new market designs and frameworks that might offer solutions.

Despite significant progress, the pace of transformation falls short of meeting global climate goals—like the international objective to triple renewables by 2030.

> Global renewable capacity is expected to grow by 2.7 times by 2030, surpassing countries' current ambitions by nearly 25%, but it still falls short of [...] the goal of tripling renewable energy capacity worldwide established by nearly 200 countries at the COP28 climate summit.
>
> — IEA Renewables 2024 Report[24]

1.5 Shifting to a decentralized, low-carbon system is monumental and needs widespread coordination across technology, policy and markets

As demand variability increases and both DERs and variable renewables multiply, more integrated coordination is key to maintaining grid stability. Electrification—driven by heat pumps, air conditioners, electric vehicles—shapes new load curves and makes them more susceptible to short-term or seasonal disruptions, especially during heatwaves.



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Figure 3. Daily average electricity demand by end use in European Union and India in the Stated Policies Scenario, 2023 and 2050[25].

Notes. GW is gigawatts, h is hour. Heating covers space and water heating; other buildings includes lighting and cooking; other includes agriculture.

Addressing these complexities requires greater system flexibility, both short-term and seasonal. Battery storage is becoming a crucial short-term dispatchable resource for managing intermittency, and demand-side measures can help reshape load patterns, shifting peak demand to periods of abundant renewable generation. Yet these solutions rely on a supportive market design, broader grid upgrades, and digital frameworks that link supply, demand, and storage seamlessly.

Digital technologies like smart meters and demand-response systems can reshape consumption, moving peak loads from evening hours to midday solar production. Nonetheless, it will take clear policies and pricing signals to make such solutions ubiquitous. A unified, cross-sector effort—integrating technical innovation, market-based mechanisms, and regulatory coordination—remains vital to building a resilient, secure system that also meets mounting energy and climate goals.

1.6 Augmenting grid infrastructure to handle digital complexity

Decentralization, digitalization, and the growing dynamism of the energy system pose both a defining challenge and a major opportunity for the coming decades. Harnessing these changes will require new market designs and coordination across diverse assets, including batteries, flexible loads, and dispatchable generation. Digital tools and software-based systems can foster real-time orchestration of distributed resources, but only if stakeholders can share data easily and trust the platforms they use.

A unified approach to policy, grid planning, and technological advancement can help us reach a sustainable energy system that supports rising demand while staying on track for climate and security goals. In the next chapter, we will see how structural barriers and a lack of coordination prevent us from fully leveraging these innovations—and why a foundational digital infrastructure may hold the key to overcoming them.

02

Structural barriers, amplified by coordination gaps, are **impeding the way forward**

The ongoing global energy transition is characterized by massive shifts toward electrification, decentralization, and digitalization. While these advances promise cleaner, more resilient, and more equitable energy systems, they also bring fresh challenges. Power grids are no longer linear, top-down structures that simply deliver electricity from a few central generators to passive consumers. Instead, they are evolving into complex, interactive networks composed of large-scale renewables, distributed energy resources (DERs), smart appliances, and numerous new participants capable of generating, storing, or trading electricity.

To fully unlock the potential of this transition, it is essential to understand the structural barriers that hinder progress, recognize the resulting coordination challenges, and explore new ways to overcome these obstacles through innovative thinking.

2.1 Structural barriers in the ecosystem today

Current solutions are purpose-specific

Many of the existing solutions address only a narrow aspect of the energy ecosystem. For example, there are platforms dedicated solely to EV charging or tools designed exclusively for DER management. Although these solutions can be effective within their limited scope, they do not provide a comprehensive framework that supports the full range of energy needs. This narrow focus means that promising innovations may only solve a single piece of the puzzle without contributing to an integrated system.

Current solutions occur in silos (non-interoperable)

Often, systems and technologies operate independently without sharing data or communicating with one another. This siloed approach leaves vital information isolated. For instance, data captured by one system may not be compatible with that from another, resulting in a fragmented landscape where opportunities for synergy are missed. Without interoperability, the full benefits of digitalization remain out of reach, and efforts to integrate new solutions are stymied.

Non-digital modes of operation

A significant portion of current processes still relies on manual or paper-based methods. These non-digital approaches lead to higher transaction costs, con-

tribute to information asymmetry, and reduce trust among stakeholders. When contracts and performance metrics must be verified through physical inspections rather than automated digital records, the process becomes both time-consuming and prone to error. Such legacy practices further complicate the scaling of innovative digital solutions.

2.2 Implications of these structural barriers

These barriers give rise to three interrelated coordination challenges –

Managing information flows

The absence of unified digital records means that data is often fragmented and inconsistent. Even in advanced economies with millions of smart meters, critical information—such as the timing and volume of energy production, consumption, and trade—remains scattered across disparate systems. This information asymmetry hampers real-time decision-making by grid operators, regulators, and market participants, sometimes leading to inaccurate load forecasts and delayed dispatch of energy.

Managing transaction flows

The reliance on manual processes and redundant documentation increases transaction costs. Without standardized digital records, **establishing trust becomes challenging—especially when traditional verification methods** (such as repeated physical inspections or manual audits) are too slow or resource-intensive. This low trust hinders market participation and delays the onboarding of new players.

Energy transactions are burdened by high costs, unstandardised processes, and limited access to low cost, dynamic, and rapid contracting mechanisms between transacting parties. Participants, such as battery owners or demand-response providers, often lack the infrastructure to independently initiate trustworthy contracts or build reputational scores. Costly digital linkages and intensive contract management further increase transaction costs, restricting the ability of individuals and small-scale energy providers to participate in energy markets without local aggregators. The absence of cohesive frameworks limits the market potential of DERs. For instance, only 25% of solar installations in the U.S. include a battery, despite the increasing availability of these technologies[26].

Managing energy flows

Without an integrated digital system to coordinate the physical movement of energy, inefficiencies and imbalances arise. These issues can lead to reduced grid stability and diminish confidence in the system's ability to handle variable renewable inputs and distributed resources effectively. Even well-designed demand-response programs may fall short if energy flows remain poorly coordinated.

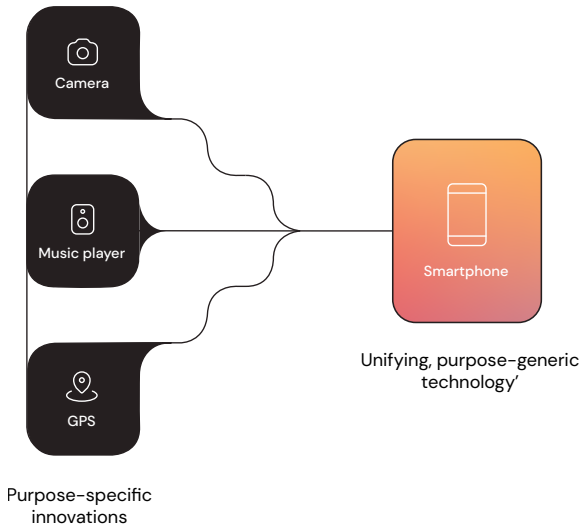
2.3 Overcoming these barriers through infrastructure thinking

Addressing these structural barriers calls for an approach that sees beyond individual projects or quick-fix policies. The concept of infrastructure-thinking with a unified approach, grounded in shared and scalable digital infrastructure, enables the energy system to move beyond piecemeal efforts and tackle the underlying lack of coordination. It is not about pushing a singular technology or enforcing a one-size-fits-all solution. Instead, it involves ensuring that minimal, open standards and protocols allow diverse actors to interconnect, share trustworthy data, and transact with reduced friction. Historical examples, such as the standardization of telephone networks, the open protocols of the internet, and uniform electrical systems, demonstrate how unification can revolutionize entire sectors.

Drawing parallels from past successes, the energy sector could adopt shared digital "rails" that facilitate interoperability among devices, networks, and administrative systems. This is seen via multiple past innovations like Pix in Brazil and UPI in India (see annexure). This approach creates a dynamic, scalable ecosystem in which new standards, tools, and security methods can be incorporated without fracturing legacy systems. It establishes a cohesive digital layer that invites every participant—from an off-grid micro-hydro unit to a cutting-edge AI-driven aggregator—to interconnect at the minimal required level and speak a common digital language. Through "programmable contracts," for instance, rules can be executed automatically once predefined conditions are met. For example, paying a homeowner when their rooftop solar sends surplus power to a neighbor's battery. These mechanisms unlock new business models, from peer-to-peer energy markets to aggregated demand-response programs, where each participant has confidence in the integrity of the data, processes, and settlements involved.

Historically, technological innovation has scaled through unification and infrastructure thinking, with purpose-specific advancements converging into broader, purpose-generic solutions. Minimal, standardized "packetization" of design and processes through open specifications has fueled rapid growth across industries—a pattern that could similarly drive efficiency, resilience, and sustainability in the energy sector.

Technological evolution has followed a consistent pattern of purpose-specific innovations converging to a purpose-generic solution

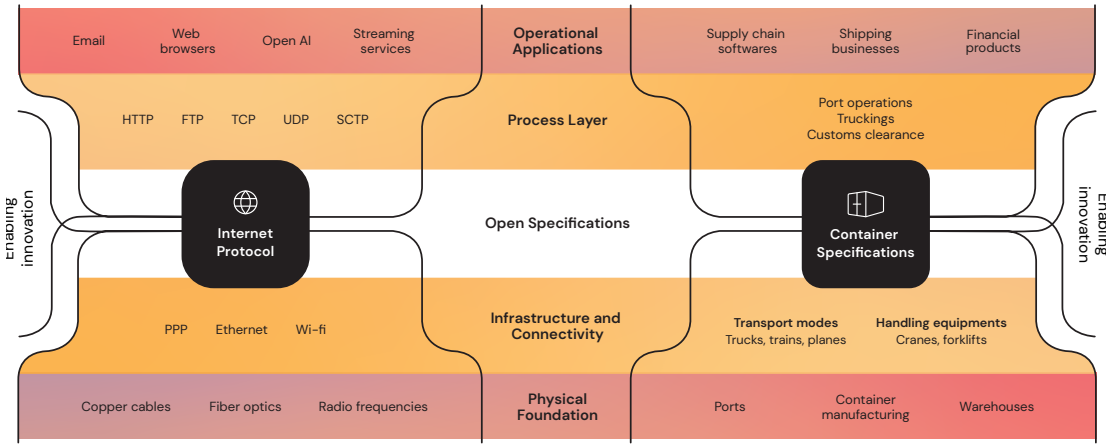


Smartphones
Unified functionalities of distinct devices (music players, GPS, cameras) into versatile tools for communication, photo editing, home automation, and more.

Cloud computing
Evolved from remote date storage to a unified system enabling data processing, machine learning and scalable innovation.

Purpose generic solutions enable seamless integration of capabilities, fostering creativity and efficiency in interconnected systems.

Minimal and generalized "packetization" of design and processes through open specifications has fueled rapid growth and scalability across industries



TCP/IP Protocol
Data packetization based on specifications allows diverse devices to exchange information seamlessly, commoditizing hardware and enabling applications like email, streaming & AI bots.

Logistics Standardization
Shipping containers revolutionized goods movement across transport modes, optimizing operations, enabling inventory management tools, and driving scalability.

Energy Sector Applications
Standardized delivery voltage, frequency, and lithium-ion battery cell formats foster interoperability, product innovation, and large-scale adoption across industries.

Figure 4. Trends of technologies that embraced unification and infrastructure thinking

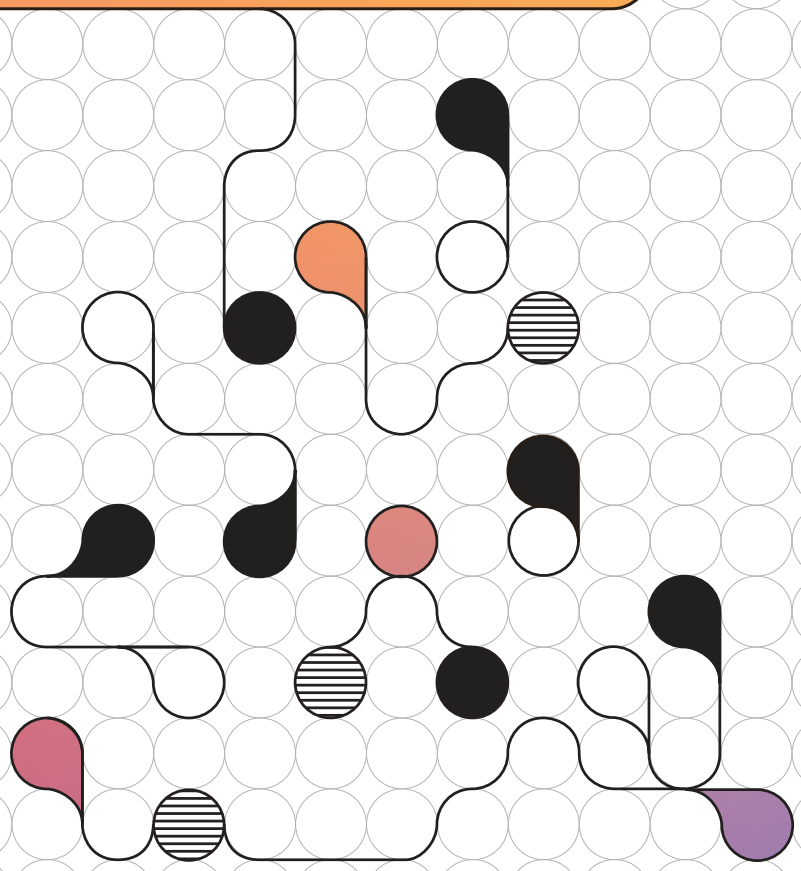
2.4
Toward an infrastructure that unlocks the next phase of the energy transition and security

With a unified energy environment in place, the energy ecosystem can transform from a patchwork of purpose-specific, siloed solutions into an integrated network that supports robust cross-sector collaboration and innovation. This transformation can reduce friction, enhance trust, and ultimately help the sector meet accelerating demand, manage variable renewable inputs, and overcome unpredictable challenges.

In short, the energy sector already has numerous promising pilots, frameworks, and technologies. What it lacks is a unifying structure to weave these efforts into a resilient, scalable fabric capable of meeting accelerating demand, variable renewables, and unpredictable shocks. The next chapter outlines how an infrastructure-led approach—embodied by the concept of the Digital Energy Grid—can shift us out of siloed thinking and into coordinated action, setting the stage for a new era of reliable, dynamic, and inclusive energy systems.

03

Digital Energy Grid (DEG) could **augment the existing energy grid** and **embody infrastructure-thinking**



The energy system grows more complex each day. Siloed fixes—ranging from specialized software platforms to targeted policy incentives—do help in particular niches, but they frequently lack the broader coordination required to address a system where everything from rooftop solar generation to demand-response programs must align. This is a problem because the next phase of the energy transition depends on integrating a growing number of technologies and participants into a unified whole. A grid that only partially sees behind-the-meter resources or that requires manual verification of countless small transactions is unlikely to keep pace with the demands of higher renewable penetration and more electrified end uses.

Infrastructure-thinking has guided the evolution of railways, telecommunications, the Internet, and digital financial services by establishing minimal yet robust common layers. These layers made entire ecosystems flourish. In the energy realm, the creation of a shared "digital architecture"—one that binds physical infrastructure with real-time data, economic transactions, and regulatory oversight—may present an equally transformative opportunity. Such an architecture can help the grid handle new inflection points by systematically knitting together the various participants, resources, and solutions under an overarching, interoperable framework.

3.1 A vision for a unified energy future

This paper introduces the Digital Energy Grid (DEG) as a conceptual blueprint for building a unified energy ecosystem. The guiding principle is to interlink the physical grid—comprising generation, distribution, loads and storage assets—with a digital "data grid" that encompasses everything from pricing signals to device identities and transaction histories. With that arrangement, we can shift from a traditional top-down network to a more adaptable, network-of-networks model, where each node (be it a microgrid, an aggregator's platform, or a prosumer's smart home) remains autonomous yet still interoperable with the rest.

By weaving together real-time data streams, operational signals, and secure exchange mechanisms, DEG sets the stage for a resilient, transparent, and innovation-friendly infrastructure. It raises the "playground" for diverse participants—utilities, regulators, communities, startups, large industries—fostering a more cohesive approach to shared goals like decarbonization, reliability, & affordability.

3.2
Key building blocks of the Digital Energy Grid

The Digital Energy Grid (DEG) centers on three interrelated elements that, when combined, enable seamless interoperability, scalability, and reliability.

i. Universal and interoperable identity for all energy assets and actors

The first cornerstone is establishing digital identities for each entity, actor and asset—ranging from power plants, transmission grids, and commercial battery farms to smaller devices like rooftop panels, electric vehicle (EV) batteries, or building automation systems. These identities act as globally unique references, akin to phone numbers or IP addresses in telecommunications.

Country codes for telephone numbers and zip codes

Each country may have its own standards for telephone numbers and zip codes, often leading to duplicates or non-interoperable formats across borders. However, the introduction of standardized prefixes, such as country codes for phone numbers (e.g., +91 for India), ensures unique identification and seamless global communication. Similarly, zip codes, while locally defined, rely on hierarchical structures that allow for clear and consistent mapping within broader postal systems. These simple yet effective standardizations demonstrate how unique identifiers can bridge regional inconsistencies, enabling global addressability and interoperability.

ii. Machine-readable data formats

Beyond simple tagging, DEG insists that metadata—about devices, energy transactions, certifications, or consumption profiles—be expressed in a way that can be systematically processed by software. Instead of static documents or PDFs, the focus shifts toward open, machine-readable templates.

Digital and machine-readable data streams

Imagine a world where every appliance's specifications, certifications, and ratings are digitally accessible in a standardized format. Combined with machine-readable consumption and transaction data, this creates the foundation for highly automated, analytics-driven, and intelligent decision-making systems. For instance, just as machine-readable invoices underpin the efficiency of modern tax systems by enabling seamless tracking and compliance, such digital streams in the energy ecosystem could unlock appliance-level metering, dynamic energy management, and precise control. This level of granularity and automation could optimize energy usage, enhance grid stability, and drive smarter energy policies.

iii. Verifiability and portability of data

The final pillar is embedding robust verification procedures, which ensure that data—once registered or recorded—is cryptographically tamper-evident and remains portable across the ecosystem. This includes cryptographic signatures, proof-of-origin records, and permissioning models that uphold data sovereignty.

Together, these three building blocks—universal identity, machine-readable structures, and verifiability—represent the minimal infrastructure necessary to align physical energy flows with the information and transactional layers of a modern, distributed system.

3.3 From grids to networks: an interconnected energy world

By layering digital identity, data interoperability, and trust mechanisms over the existing physical grid, DEG transforms today's somewhat siloed landscape into a more interconnected "network of networks." In practical terms, it means that:

- Transmission and Distribution Operators can have real-time, programmable access to each distributed asset (be it an electric vehicle [EV] or a rooftop photovoltaic [PV] array), enabling more intelligent load dispatch and grid-balancing decisions.

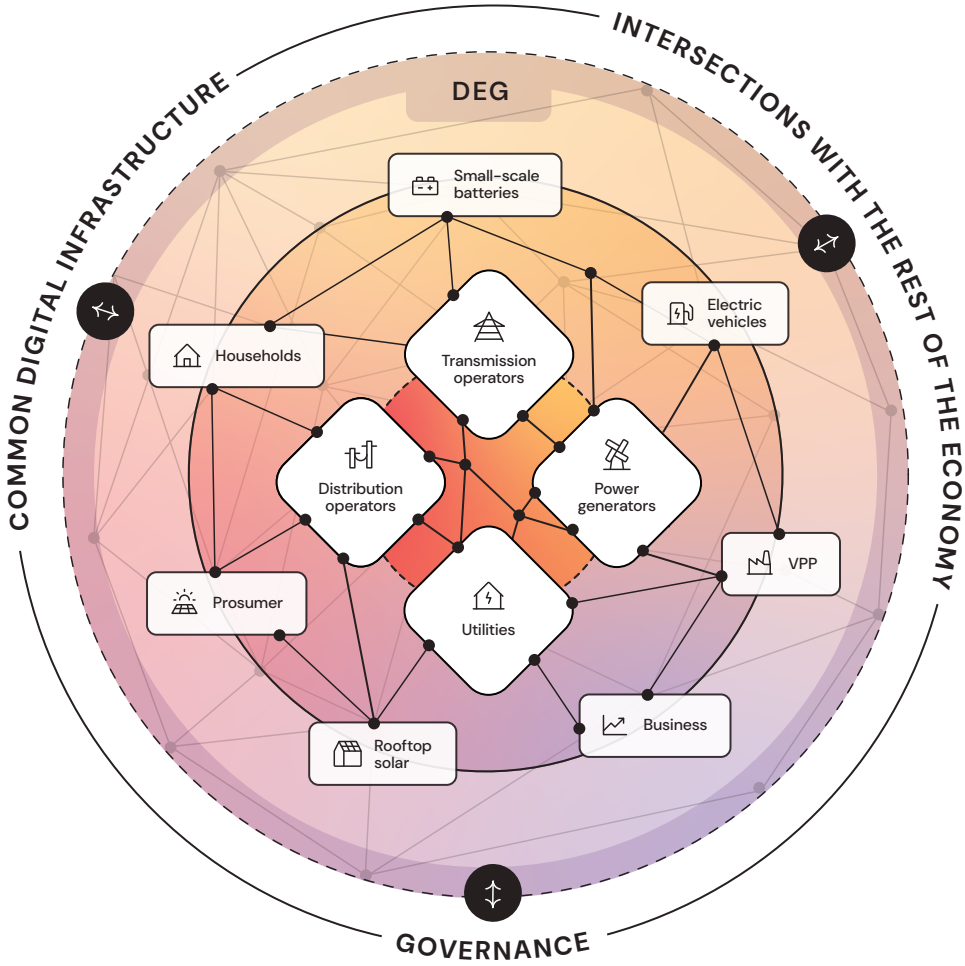


Figure 5. The DEG vision encompasses all facets of the ecosystem.

- Energy Producers can automate everything from certifications (like proof of renewable generation) to contract settlements, while verifying their performance with cryptographic signatures.

- Consumers gain new freedoms, such as frictionless switching to different suppliers or direct peer-to-peer transactions. Households might securely lease out unused battery capacity or easily join local microgrids for resiliency.

- Regulators and policymakers can embed oversight directly into the infrastructure, ensuring that safety, data privacy, and consumer protections scale with the network. They also gain clearer visibility into real-time grid conditions, improving policy effectiveness.

3.4
Bridging silos to capture combinatorial value

Digital solutions, open standards, and advanced technologies like OCPP, OCPI, IEC 61850, and initiatives such as Project Mercury[27] are already enabling targeted interoperability and communication for EV charging, DER management, and clean technology integration. Building on this progress, these efforts have successfully addressed specific needs, yet there remains an opportunity to unlock even greater potential through universal composability across the energy grid. A general-purpose foundation can amplify their impact significantly.

The Digital Energy Grid (DEG) can coordinate, complement, and unify existing solutions-without displacing them. By enabling true interoperability across systems, DEG provides the broader architecture on which these purpose-specific standards can be layered and expanded.

Have we seen this before?

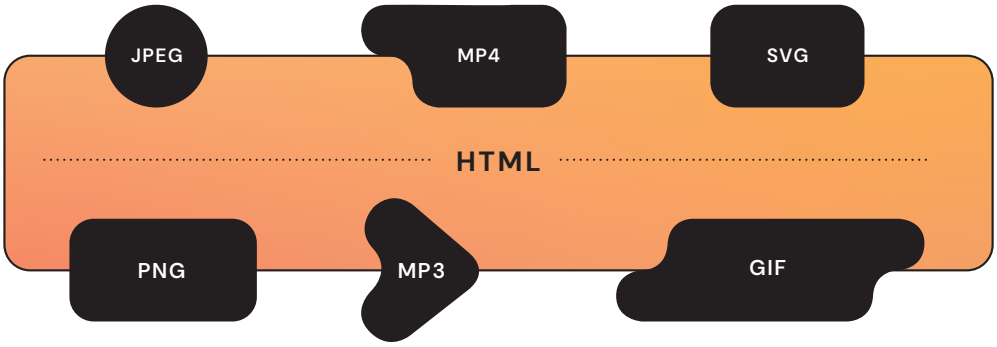


Figure 6. General purpose standard accommodates and amplifies purpose-specific efforts.

A universal base layer can integrate and unify various purpose-specific standards, enabling seamless interactions & creating combinatorial value.

For example, JPEG, GIF, SVG, MP4, & MP3 are purpose-specific standards, each designed for a distinct data format- JPEG for still pictures, GIF for animations, MP3 for audio, and MP4 for video. While each excels in its function, they cannot inherently interact with one another. This is where a unifying base layer, like HTML and HTTP, comes in. HTML allows different types of content – images, videos, text, and audio – to coexist and function together within a single webpage, while HTTP facilitates the smooth transfer of this multi-modal data across networks.

If purpose-specific standards are like individual Lego blocks of different shapes and sizes, the unifying base layer is the set of connectors and the instruction guide that allows these blocks to fit together in meaningful ways. HTML and HTTP act as the underlying fabric within which diverse elements can interoperate, empowering innovators to create rich, interactive, and multi-modal experiences that seamlessly combine different formats into something greater than the sum of their parts.

Instead of creating competing standards that replace existing frameworks, a well-architected base layer enables existing standards to work together cohesively. This composability- the ability to combine different components into larger, functional systems- ensures long-term adaptability. The base layer remains future-proof, allowing new purpose-specific standards to be integrated without requiring a complete overhaul of existing systems.

3.5 DEG as an evolutionary vision, not a single product

It's important to note that DEG is not a fixed piece of software or a proprietary technology stack. Instead, it is a vision for an evolving, interoperable, and modular set of digital rails spanning the entire energy sector. Over time, the ecosystem can accommodate new standards, tools, and security methods without fracturing. Legacy systems can gradually integrate, while advanced players adopt next-generation functionalities for even deeper automation or cross-sector linkages.

Here, we see infrastructure-thinking in action: the objective is not to enforce uniformity by discarding existing systems, but to extend an invitation for each actor to interconnect at the minimal required level. Just as the internet did not demand that all computers be made by the same manufacturer, DEG does not demand a universal software or hardware solution. Instead, it sets the stage so that all participants—from an off-grid micro-hydro unit to a cutting-edge AI-driven aggregator—can speak a shared digital language and produce verifiable records.

The next chapters examine how DEG can unlock new possibilities across different parts of the energy ecosystem and how this vision, though ambitious, can be realized through incremental actions and collaborative governance. By focusing on the foundational digital building blocks today, we can ensure that energy systems worldwide can adapt, innovate, and collaborate on a scale never before seen—meeting both current demands and the pressing need for a sustainable future.

04

Unlocking **new possibilities** with **DEG**

A system founded on universal identity, machine-readable data, verifiability, and portability does more than simply organize information; it also enables transaction logic that can be automated, combined, and adapted. When energy assets and participants are all "on the grid" in a digital sense, functions like peer-to-peer trading, dynamic pricing, and transparent certification can emerge without sacrificing the oversight that regulators, auditors, and financial institutions require. By weaving these capabilities into a common infrastructure, the Digital Energy Grid (DEG) sets the stage for more interconnected and imaginative solutions across the energy ecosystem.

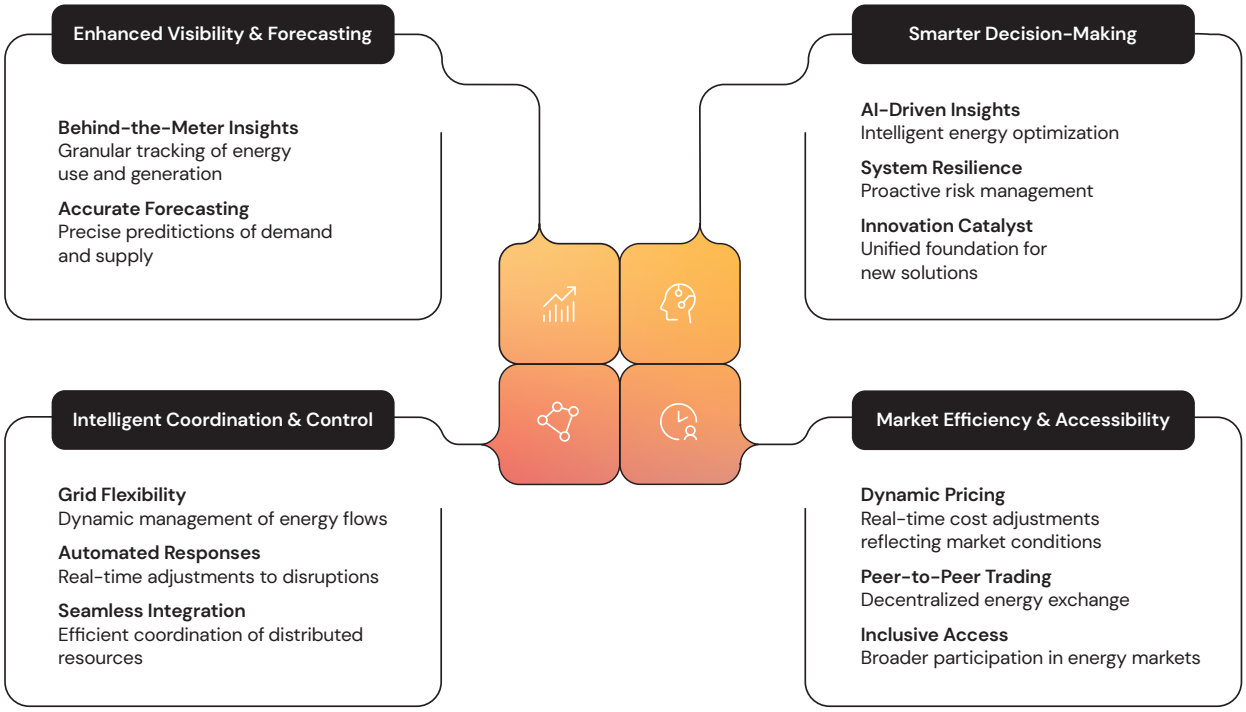


Figure 7. DEG combines transparency, innovation, and dynamic market solutions.

4.1 Illustrating the value: Different stakeholders, different needs

DEG offers wide-ranging gains for individuals, businesses, governments, and the energy sector at large. The table below highlights how various stakeholders in the energy ecosystem might experience the transformation once they integrate with a coherent digital network built on the principles of universal identity, machine-readable data, and verifiable records.

As a...	What am I looking for...	What will I get through DEG...
Consumer/Prosumer Residential: Individual, household, apartment, gated community, cooperative housing society Commercial: Corporation, government agency, SMEs, industry, factory, township, nonprofit Energy Cooperatives: Local collectives managing shared energy resources	• Reliable and affordable energy • Transparency in billing and usage data • Path to self-generate or store power • Incentives for efficiency or renewable investments • Control over personal data • Enhanced energy security and resilience • Reduced carbon footprint and verified access to renewables	• **Immediate ways** to buy surplus energy or sell rooftop solar output at peak prices, with **automated billing** and **real-time price signals** • **One-stop discovery** of applicable subsidies or financing programs, **digitally verified** for quicker approvals • **Secure, user-managed data sharing** with banks or insurers to prove capacity, consumption, or production—**speeding up** loan or claim processing • **Fractional ownership**[28] in community renewables, tracked **on a shared dashboard** with **digitally enforced rights** and **transparent payouts**
Utility Company / Grid Operator National or regional operator, electricity retailer, microgrid or battery operator	• Real-time visibility into supply and load • Ways to balance the system and ensure reliability. • Reduced congestion and transmission losses. • Integration of renewables and DERs without destabilizing the grid. • Streamlined processes for maintenance and investment planning.	• **A unified, real-time view** of all distributed assets—simplifying peak load management and **accelerating** dispatch decisions • **Automated onboarding** for new producers or EV fleets, with **pre-verified** digital credentials that cut overhead and **reduce errors** • **Fine-grained forecasting** that combines consumer patterns, weather data, and aggregator signals for **precise** short- and long-term load balancing • **Streamlined maintenance** and asset planning, thanks to **verifiable operational logs** shared across multiple stakeholders
Energy Producer Central power plant, renewable IPP, VPP operator, aggregator	• Maximized revenue from energy production • Quicker grid interconnections • Access to dynamic markets and carbon credits • Forecastability and streamlined operations	• **Faster regulatory approvals** for new projects, via **machine-readable** credentials recognized by local authorities and financiers. • **Decreased curtailment**, since operators can match supply with **flexible loads** in near real time—improving ROI on generation assets. • **Transparent performance data** that proves renewable credentials for

As a...	What am I looking for...	What will I get through DEG...
	• Ability to demonstrate compliance and performance data.	carbon-offset trading or compliance, **instantly recognized** by global markets • **Prompt settlement** for energy delivered, triggered by **verified production logs** automatically shared with buyers and regulators
Energy Market Operator Independent system operator, transmission system operator, or electricity exchange	• Efficient, transparent trading mechanisms' • Management of supply and demand volatility • Broader participation in energy markets • Low transaction costs and faster settlement • Growth in advanced dynamic pricing and demand-response models	• **Real-time bids** from thousands (or even millions) of small producers and prosumers—creating **liquid, highly competitive** markets • **Automated credential checks** so new DER participants can join quickly, while ensuring **full compliance** with market rules • **Dynamic tools** to reduce curtailment by pairing surplus power with flexible demand, supported by **verifiable transactions** • **Accurate pricing** of ancillary services, capturing micro-level flexibility from consumer devices and distributed storage
Regulator State or national energy agency, local authority, energy policy institution	• Universal energy access • Meeting national and international climate targets • Safeguarding safety, resilience, and affordability • Fair and transparent markets • Consumer protection and privacy • Incentivizing innovation	• **Richer, real-time data** on how resources are used and where they're lacking—supporting **evidence-based** policies and **targeted** subsidies • **Faster, more cost-effective** enforcement of consumer protections or grid codes through **automated compliance** checks and alerts • **Automatic tracking** of emissions, renewable energy credits, or carbon offsets—**streamlining** climate reporting and policy updates • **Digitally encoded** tariffs, safety standards, and incentives that **immediately propagate** to all energy participants, **cutting red tape**
Financing agency Development bank, commercial lender,	• Transparent risk-return profiles for energy projects	• **Tamper-proof** performance logs and asset credentials, ensuring **faster, lower-risk** due diligence on energy projects

As a...	What am I looking for...	What will I get through DEG...
insurance company, microfinance institution	• Verification of asset performance and viability • Trustworthy channels for small-scale projects (e.g., microgrids, local solar) • ESG alignment and sustainable finance	• **Simple onboarding** of micro-energy investment categories (community solar, aggregated storage) with **transparent operational data** • **Real-time event triggers** (e.g., battery downtime) that **automatically release** insurance payouts or loan tranches upon verified conditions • **Enhanced credit** evaluations and ESG scoring, by analyzing **consistent data** on project viability and asset utilization
Market Players Charge point operators, battery manufacturers, appliance makers, AI solutions providers, data center operators, industrial economic zone oeprators	• Expansion of new offerings and revenue streams • Efficient integration with grid operations for affordable and sustainable energy sources • Compatibility with emerging standards and systems • Access to funding for R&D	• **Easier multi-utility integration**, thanks to **universal data formats** and digital asset IDs—**reducing** cross-platform overhead • A **bigger, more uniform** customer base, with fewer barriers to launching new services (e.g., EaaS) across multiple jurisdictions • **Seamless collaboration** with grid operators, as new hardware or software can be **digitally verified** for compliance automatically • **Lower transaction costs** and **quicker product rollouts**, enabling more specialized, high-value offerings that complement existing platforms
City Administrators Municipal authorities, urban planning, transportation boards	• Reduced carbon footprint and improved public services • Reliable local power generation and distribution • Economic growth through local energy initiatives • Tools for tracking progress toward sustainability goals	• Accelerate **smart city's net zero plan** that coordinates street lighting, electrification of vehicles, and community solar under **one digital umbrella**—all tracked in **real time** • **Targeted incentives** for local energy initiatives, creating **green jobs** (e.g., system installers, energy auditors) and **shared ownership** of energy assets • **Granular data** on power usage and emissions for each neighborhood—**enabling** more efficient urban planning and **faster** emergency response • **Automated policy** and rule enforcement, with **digital dashboards** showing progress toward local sustainability or resilience goals

As a...	What am I looking for...	What will I get through DEG...
Government National government, diplomatic entity	• Affordable, stable energy supply for citizens • Achievement of climate commitments • Reduced dependence on imported fuels • Scalable, modern grids • Encouragement of private sector participation	• **Streamlined interactions** between federal agencies, utilities, and local entities—**accelerating** major infrastructure projects with consistent data • **Real-time metrics** on policy effectiveness, enabling **flexible** adjustments to electrification or decarbonization targets without bureaucratic delays • **Attractiveness for foreign** and domestic investment, since **open, verifiable data** and universal identities reduce risk and boost investor confidence • **Scalable, modern** grid policies that adapt to diverse regions and resource types, ensuring stable, climate-aligned energy for citizens

Table 1. Stakeholders can adopt different facets of the DEG Vision.

4.2 A foundation for synergistic growth

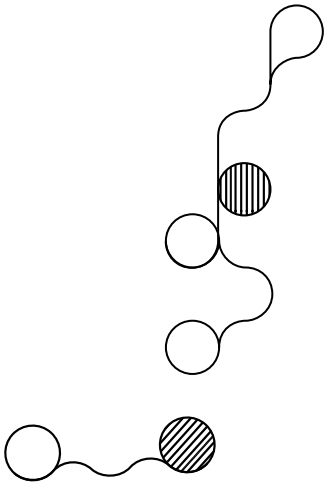
Each type of stakeholder—consumer, producer, operator, regulator, financing agency—can adopt DEG at a different scale, reflecting local conditions or immediate priorities. Yet as they link into the same underlying network, their combined actions create a synergistic whole that is more robust, transparent, and ready to evolve over time. Large utilities may be the first to standardize identities for their generating units; local communities might join soon after, using the same protocols to unlock real-time data on rooftop solar. Little by little, segments of the energy ecosystem that used to operate in isolation merge into a more unified environment.

At a macro level, this transformation carries meaningful implications for clean energy adoption, grid stability, and the everyday experience of energy users. Overcoming the longstanding fragmentation in how data is managed, trust is established, and transactions are settled could deliver compounding benefits—a better match of supply and demand, deeper cost savings, and less friction in scaling up new infrastructure.

4.3
Core functionalities for transformative scale

DEG weaves core functionalities, such as observability and secure transactions, directly into the infrastructure. With universal identity, machine-readable formats, and cryptographic assurance, the physical flow of electrons can be coordinated with seamless, digital interactions. A flexible marketplace emerges, fostering collaboration and innovation at scale while preserving reliable oversight and guaranteed fairness for all. In place of patchwork pilots or proprietary "smart grid" experiments, the industry gains a versatile digital framework that encourages both public and private stakeholders to expand, refine, and recombine capabilities.

This approach sets the stage for the next chapters, which illustrate the breadth of possibilities DEG can unlock, from frictionless subsidy distribution to advanced peer-to-peer trading and cloud battery services. By creating a single environment where devices, participants, and transactions can interoperate, DEG aims to reduce wasted effort, open new revenue opportunities, and realize an energy transition that lives up to its transformative promises.

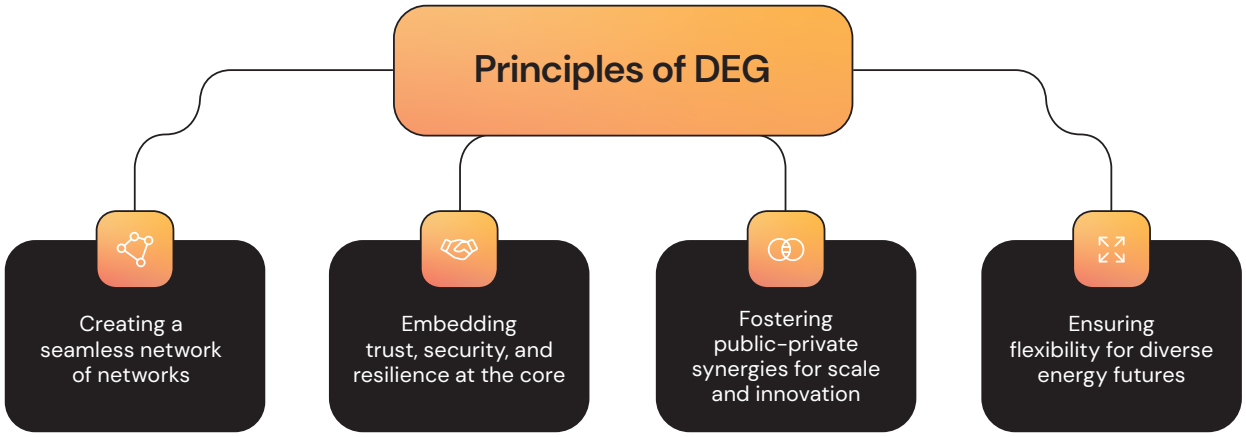


05

The **DEG vision embodies scale, diversity, openness,** and **resilience** for inclusive energy systems

The Digital Energy Grid (DEG) serves as more than a conceptual framework: it offers a vision of an inclusive, global energy ecosystem, one that brings together various technologies, participants, and use cases. Realizing this vision depends on weaving essential design principles into every level of the grid. These principles—openness, trust, resilience, public-private synergy, and flexibility—are not abstract ideals but rather practical necessities that shape how DEG is structured, governed, and adopted across different contexts.

While DEG is a powerful example of infrastructure thinking, it is only one of several possible approaches to guide the energy transition. Our focus here is on how infrastructure-led methods can make the system more robust and adaptable, without prescribing a particular approach.



5.1 Creating a seamless network of networks

A key ambition for DEG is to function as a "network of networks," unifying distributed energy resources (DERs), IoT-based devices, and energy infrastructures spanning localities and national borders. This allows seamless interaction among diverse systems, regardless of origin or design. For example, a city's community battery system can coordinate with a regional grid operator or an industrial microgrid can share data with local regulators, all via the same coherent network.

Interoperability as a core principle

Interoperability goes beyond simply enabling communication; it ensures that each subsystem—whether an EV charging network, a micro-hydro plant, or a rooftop solar aggregator—uses minimal, open, shared digital standards. This cuts the cost of "translating" data and simplifies how new entrants tap into markets or compliance structures.

Modular design allows anyone to build on the network

By unbundling problems and solutions into core, reusable building blocks with open protocols and specifications, we can ensure that these components are highly trusted and cost-effective for reuse by public and private entities. This approach allows the ecosystem to combine these modular building blocks, much like Lego pieces, to create tailored solutions that fit various needs.

By adopting a minimalist approach, the architecture is designed to excel at a single function without imposing on or disrupting the rest of the workflow.

Minimizing unnecessary fragmentation

Instead of letting each entity build closed, proprietary systems, a widely shared "digital backbone" reduces fragmentation at scale. It eliminates the repetition of efforts —including identification, verification, and other forms of data reconciliation—thereby enabling seamless interactions across the system, breaking siloes, and unlocking combinatorial benefits.

Design for evolution and adaptation

Technological progress can take place quickly, and rigid platforms risk becoming obsolete. DEG's open standards and protocols are designed for modular upgrade paths, allowing new solutions to plug in or swap out without breaking existing connections. This future-readiness lets participants upgrade hardware or adopt new software without disrupting the entire system, thus preserving backward compatibility and reducing disruption.



5.2 Embedding trust, security, and resilience at the core

As millions of DERs join the energy grid, the challenges of ensuring reliability and security multiply. Cyber threats, large-scale privacy breaches, or malicious activities can threaten confidence in the ecosystem. In a decentralized environment, the stakes become even higher—one vulnerability in a microgrid or connected appliance could cascade across the system. DEG has been designed keeping these stakes in mind.

Verification and cryptographic integrity

Verifiable data, identity systems, and cryptographic proofs are woven into the design to guarantee that no node can falsify or tamper with asset metadata, transactions, or operational logs without being detected. This is vital for maintaining trust in transactions like peer-to-peer trading or flexible load contracts.

Security by default

Rather than bolting on security after deployment, the DEG framework treats security and privacy as foundational requirements. For instance, standardized protocols can specify cryptographic signatures for every data packet, ensuring that private data is encrypted end-to-end and that only authorized parties can access, authenticate or alter it with consent.

Resilience and redundancy

A robust system is one that can bounce back from localized failures. DEG's decentralized network architecture is designed to avoid single points of failure by enabling different players and solutions to plug in when needed, and to incorporate redundancy in data storage, node operation, and validation. This reduces the likelihood that a local outage or security event spirals into a major, system-wide disruption.



5.3 Fostering public-private synergies for scale and innovation

Building a digital energy infrastructure that spans countries and continents requires broad collaboration among governments, regulators, utilities, technology developers, and grassroots innovators. While public institutions lay down essential guardrails, private entities bring capital, expertise, and adaptability that can accelerate widespread adoption. Co-creation through pilot projects and shared frameworks can unite these strengths: for instance, a local aggregator might partner with a national utility to test real-time price signals or plug-and-play solar kits, with results feeding back into broader policy and market designs.

This also opens doors for innovation across the system, by different kinds of participants to solve a diverse range of problems in the ecosystem. By lowering barriers for smaller players or specialized startups, synergy can extend solutions to underserved areas, ensuring that no segment of the energy community is left behind. Over time, success stories from local experiments can inspire new entrants in other regions, creating a positive feedback loop that drives further improvement.



5.4 Ensuring flexibility for diverse energy futures

The energy landscape varies from one context to another. Megacities may emphasize rapid EV adoption and large-scale renewables, while smaller towns might concentrate on community-level microgrids or shared solar. DEG vision acknowledges this diversity by enabling "modular" adoption of its core elements, adjusted to local conditions and readiness levels.

Modular, asynchronous approaches

One region could begin by assigning unique digital identities to each rooftop solar installation, while another might prioritize data structures for EV charging stations or advanced grid automation. Each domain can adopt the portions of the DEG framework that address its immediate challenges without losing compatibility with future expansions.

Equitable access and inclusivity

A universal framework means that the same open protocols and identity systems can be used by large utilities and small local energy cooperatives alike. This levels the playing field, so even resource-constrained communities can tap into modern grid functionalities—like demand response or dynamic tariffs—and benefit from a shared network.

Adapting to emerging technologies

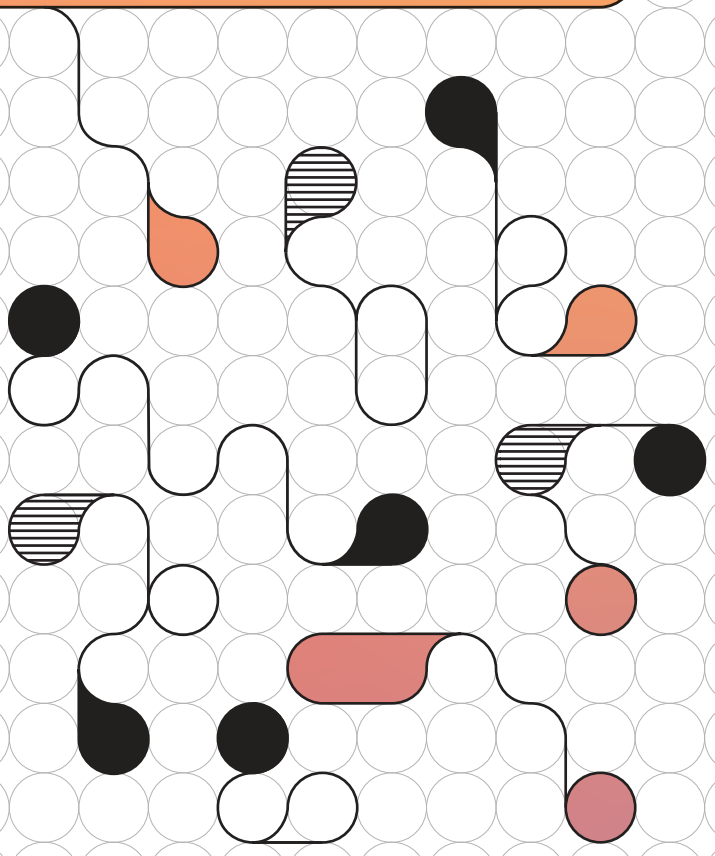
Flexible design also ensures that DEG can incorporate new capabilities. Whether that's hydrogen-based microgrids, AI agents scheduling load in real time, or advanced carbon-trading algorithms, the grid can keep evolving in ways that respect local differences while preserving overarching interoperability.

By incorporating scale, diversity, openness, and resilience as core design pillars, the Digital Energy Grid aims to create an environment in which participants—large or small—can thrive. Openness frees them to experiment, trust mechanisms help them transact and collaborate safely, and well-defined governance structures support further expansion and innovation. The outcome is not just a patchwork of discrete fixes, but a comprehensive ecosystem that helps societies everywhere pursue a secure, inclusive, and climate-aligned energy future.

These same principles animate the examples explored in the next chapter, where today's projects and tomorrow's innovations illustrate how a unified grid can advance the transition from fragmented, siloed approaches to a flourishing, resilient energy network.

06

Through coordination, **DEG** can **enable** a **wide range** of **current** and **future possibilities**



The potential applications of a Digital Energy Grid (DEG) span far beyond those we can foresee today. A unified infrastructure for energy flows, information exchange, and secure transactions could lead to breakthrough innovations that resemble the chain reactions triggered by the emergence of the internet. Where connectivity and interoperability become the norm, unforeseen "flywheel effects" often follow—akin to how the web eventually paved the way for social media platforms, the gig economy, and generative AI.

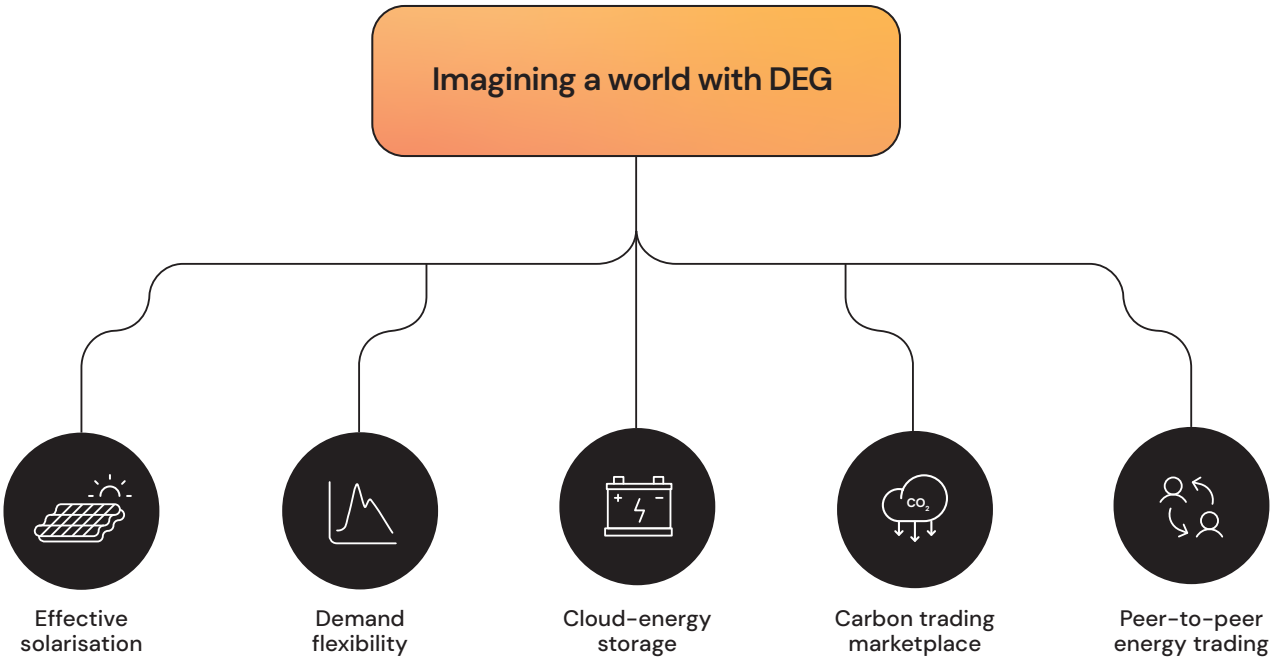


Figure 8.
Imagining a world with DEG.

Note.
All imaginations are hypothetical. Logos used are for illustrative purposes only. All names are ficticious, and any resemblance is coincidental.

The examples in this chapter offer a glimpse into the wide variety of applications and the concrete value DEG can bring to both existing and nascent fields. These hypothetical scenarios are not exhaustive; rather, they illustrate what becomes possible when energy assets, participants, and transactions operate under common, interoperable standards.

6.1
Frictionless subsidy distribution and effective solarisation

In a remote village in a country with large electricity use in agricultural pumping, the regional government is launching a solarisation initiative to accelerate rooftop solar adoption. The goal is to shift households from dependency on highly subsidised free electricity to becoming self-sufficient energy producers. Through this program, families like **Mihir's**, a local farmer, are eligible for a 50% subsidy on 3kW rooftop solar panels, helping them reduce their reliance on grid power.

However, under the current setup, Mihir's path to solar adoption is a maze of disjointed applications and verifications. He applies at three different places for subsidy, loan (for the rest of capital) and net-metering connection to his energy retailer. The lack of trust in the subsidy ecosystem prompts Mihir to submit proof of identity, property ownership, and eligibility, only to face separate applications for each. Even after securing approvals, he buys the panels, submits proof of purchase, ensures they meet specification requirements, and confirms installation with a certified service provider.

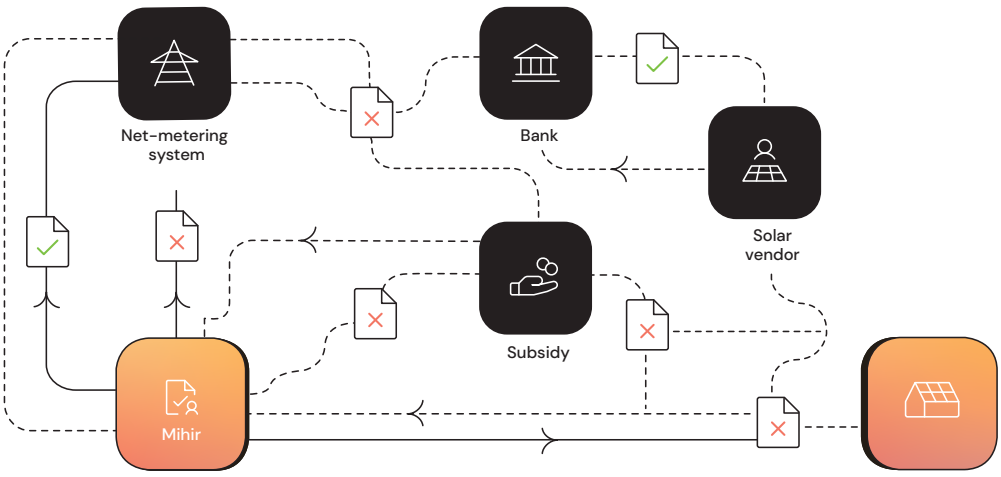


Figure 9.1
Illustrative diagram of solar subsidy distribution workflow in a non-DEG environment

This is an administrative burden for Mihir and a trust challenge for the government, as the absence of real-time verifications slows approvals. To complicate matters, the bank that finances Mihir's solar project needs to see his subsidy eligibility and pre-approval status before disbursing the loan—once again in a siloed, manual process.

This is where the Digital Energy Grid (DEG) comes in. Built on common specifications and standards, DEG brings the government, banks, utilities, and service providers into one cohesive digital ecosystem. Mihir's subsidy application becomes a single streamlined process—eligibility, approvals, and financing all tied together. When he applies for financing, the bank seamlessly verifies his pre-approved subsidy status. Certified vendors and skilled installers, each with verifiable digital credentials, are discovered and contracted with minimal hassle.

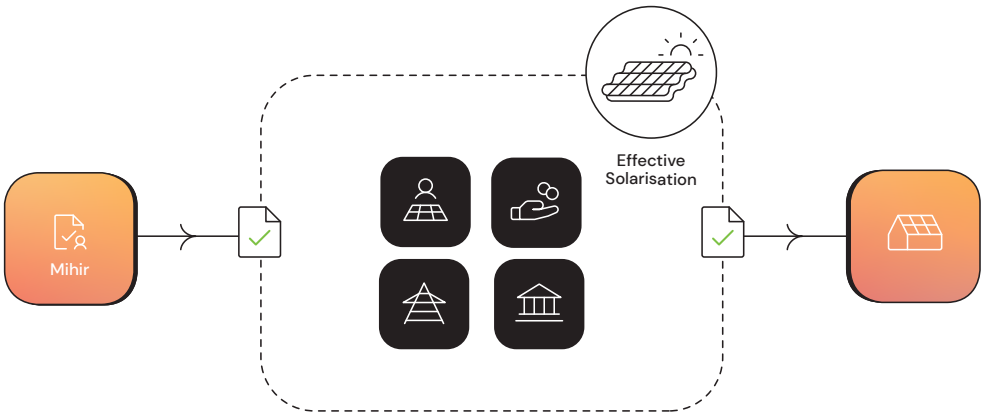


Figure 9.2
Illustrative diagram of efficient solar subsidy distribution workflow in a DEG environment

With DEG, real-time telemetry from Mihir's new solar system is accessible, with Mihir's digital consent, to both the government and the bank, ensuring transparency in performance. Once the system is verified as active, his subsidy is automatically disbursed, and the loan terms adjust accordingly. The siloed steps collapse into a frictionless, auditable flow.

By removing administrative roadblocks, DEG merges finance, subsidies, and energy data, cutting delays and reducing costs. It connects all key actors on a single interface, so a process that used to be fraught with paperwork, repeat verifications, and mistrust now flows seamlessly, giving Mihir and thousands like him a straightforward path to going solar.

6.2
Smart grid resilience: Enhancing demand response programs

Paul, a grid operator for a city with high penetration of EVs and air conditioning loads, monitors rising electricity demand as summer heat peaks. Tasked with maintaining grid stability, Paul partners with a platform called Flex to run a citywide demand response (DR) program.

Emma, a consumer in Paul's city, uses a bill-payment app called TapPay. Through TapPay, she's enrolled in a program that allows her air conditioner and EV charger to respond to grid signals. On scorching days when electricity demand soars, Paul's platform automatically sends a message via TapPay, offering Emma a small rebate if she allows her thermostat to adjust by a degree or two. Emma taps "accept," and her thermostat dials back slightly—she barely notices. Thousands of other consumers do the same. This aggregate demand drop relieves strain on Paul's network, preventing brownouts. In return, the city meets its peak needs without resorting to pricier generation options.

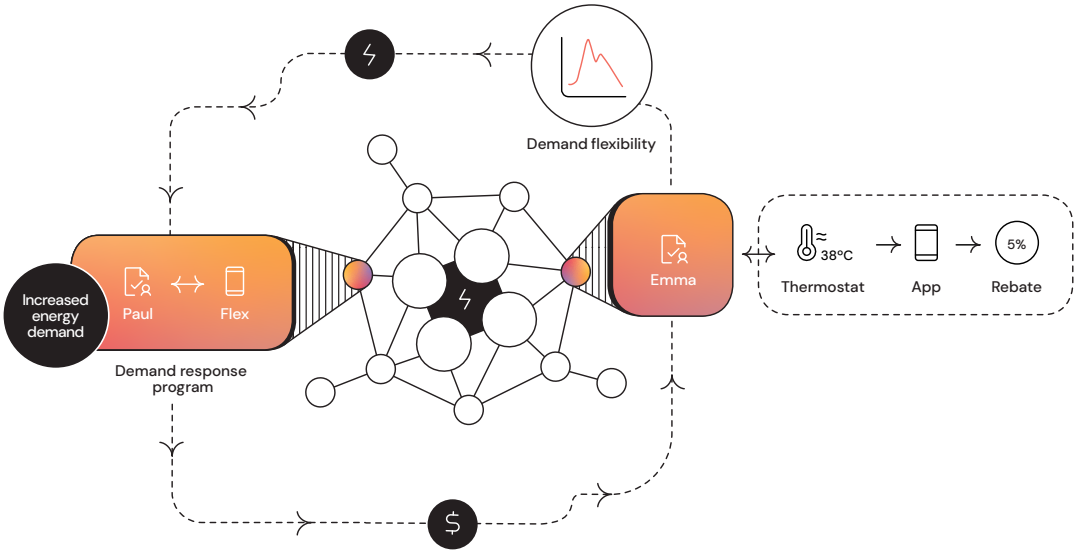


Figure 10. Conceptual example of demand response program flow using DEG-enabled coordination.

Later, settlement is equally seamless. Flex aggregates how much load was shed, verifying each household's response through real-time, cryptographically signed data from the city's smart meters. Emma's discount is automatically credited to her TapPay account, and Paul sees a neat, verifiable record of who participated and by how much. In this way, the Digital Energy Grid fosters rapid coordination among operators, aggregators, and consumers. It unifies the necessary data streams—meter readings, price signals, device statuses—into a single trustworthy system, boosting grid resilience and rewarding everyday consumers for contributing to stability.

6.3
Cloud energy storage: Unifying distributed assets

David, a resource manager at a major utility, faces a steep evening peak on hot days. Historically, the utility might have built expensive grid-scale battery systems or fired up peaker plants to meet demand surges. But with the Digital Energy Grid, David has a new option: renting spare storage capacity from households and businesses.

Sophia is a homeowner in the utility's service area. She recently installed a behind-the-meter battery to complement her rooftop solar. Through a platform called **PowerLink**—which connects to DEG—Sophia can "lease out" part of her battery's unused capacity. She lists her availability and price on the platform, visible to David at the utility. On a sweltering evening, David's monitoring system flags a sharp peak forecast. He logs onto PowerLink, sees hundreds of batteries like Sophia's offering capacity, and instantly dispatches them to feed stored energy back into the grid for a few hours. Each household or business battery discharges slightly, collectively meeting the peak without requiring new infrastructure.

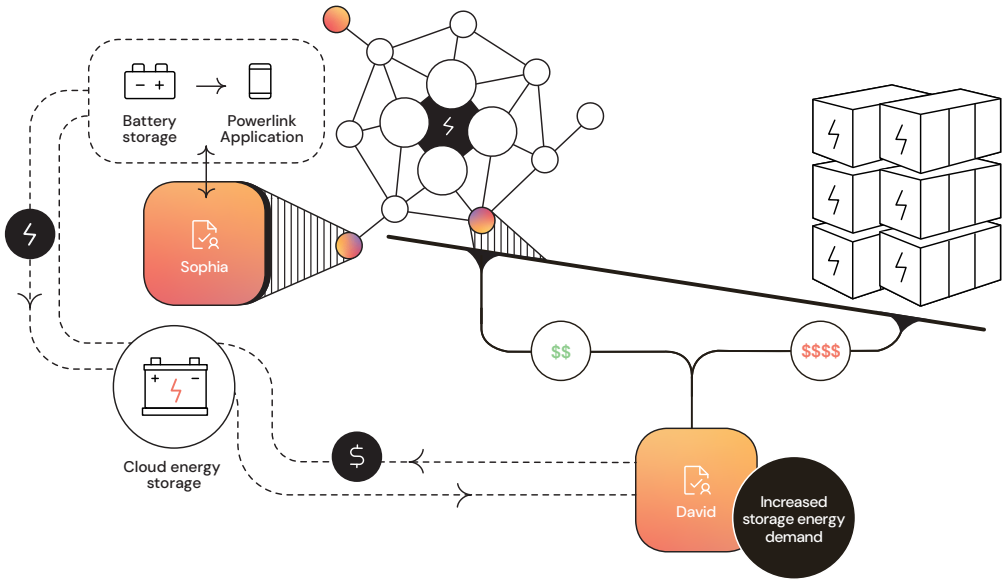


Figure 11. Illustrative depiction of unifying energy assets through cloud energy storage.

Thanks to universal identity and verifiable data, the utility confirms each battery's state of charge and disbursement timeline. Once the event ends, David's system automatically compensates every participant. Sophia notices a credit in her energy wallet—her home remains powered, and she's earned money by helping the grid. The result? Lower capital costs for the utility, less strain on the grid, and an empowered community of prosumers.

6.4
Carbon trading innovation: A global carbon marketplace

Sarah, an environmental compliance officer at a manufacturing firm, is racing to meet strict carbon reduction targets. Her company opts to purchase verified carbon credits to balance out residual emissions.

Meanwhile, **Amit** manages a solar farm in another part of the world, generating surplus green electricity and carbon offsets. Under conventional systems, Amit's credits might be locked to a local registry or require cumbersome cross-border verifications.

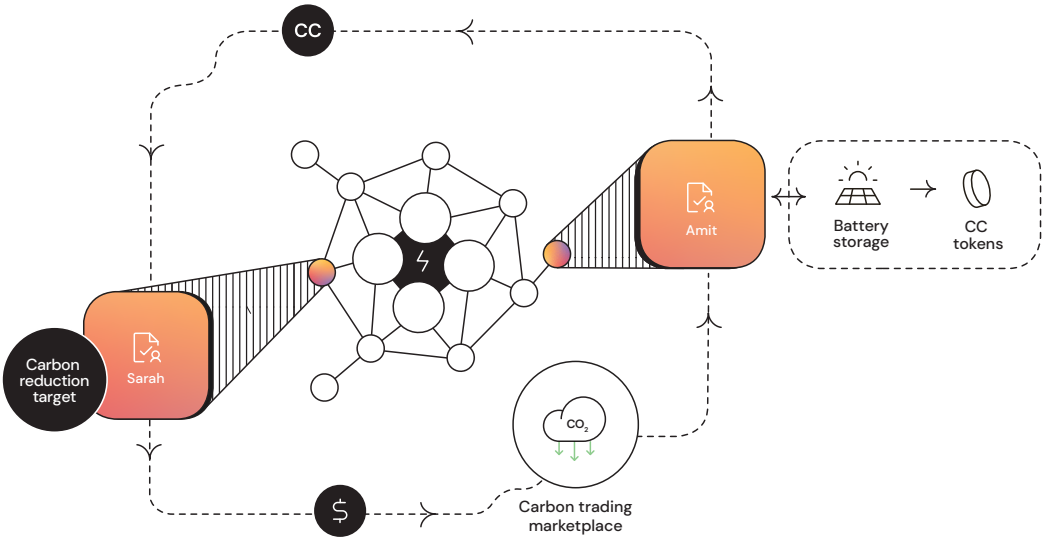


Figure 12. Illustrative depiction of carbon credit tokenization and trading facilitated by DEG.

With DEG, Amit's solar farm is registered with a universal, verifiable identity. Every megawatt-hour it produces is securely logged, generating cryptographically verifiable carbon credits. Sarah's firm, located halfway across the globe, can discover Amit's credits on a global carbon marketplace integrated into DEG.

When Sarah's company purchases these credits, the transaction and subsequent retirement of credits are automatically recorded. Amit sees immediate payment, while regulators on both sides trust the cryptographic audit trail. The once-complex cross-border carbon trade now flows seamlessly—no double counting, no paper friction, and real proof of offset.

6.5
Vehicle-to-grid innovation: Smart EV charging initiative

Hans, a forward-thinking regulator in a developed country, sees electric vehicles (EVs) as more than just energy consumers; they're potential grid resources. Working with local distribution companies, Hans launches a vehicle-to-grid (V2G) policy pilot.

Irene, a researcher at an energy innovation lab, uses DEG to test new V2G technologies. She enrolls volunteers like **Thomas**, an EV owner, who charges his car overnight. Because Thomas's EV is recognized by a universal identity in DEG, his vehicle can securely share real-time battery info.

On hot afternoons, the lab's system signals Thomas's car to discharge a portion of its battery back to the grid for an hour. In exchange, Thomas earns credits on his energy bill. **Simone**, another EV user, times her charging to 2:00 a.m. when wind power is abundant. The entire process is orchestrated seamlessly through cryptographically verifiable data flows, so Hans's regulatory office can track how much capacity is injected back into the grid.

At scale, this V2G approach turns thousands of EVs into mini power plants—stabilizing the grid, cutting costs, and accelerating the clean energy transition.

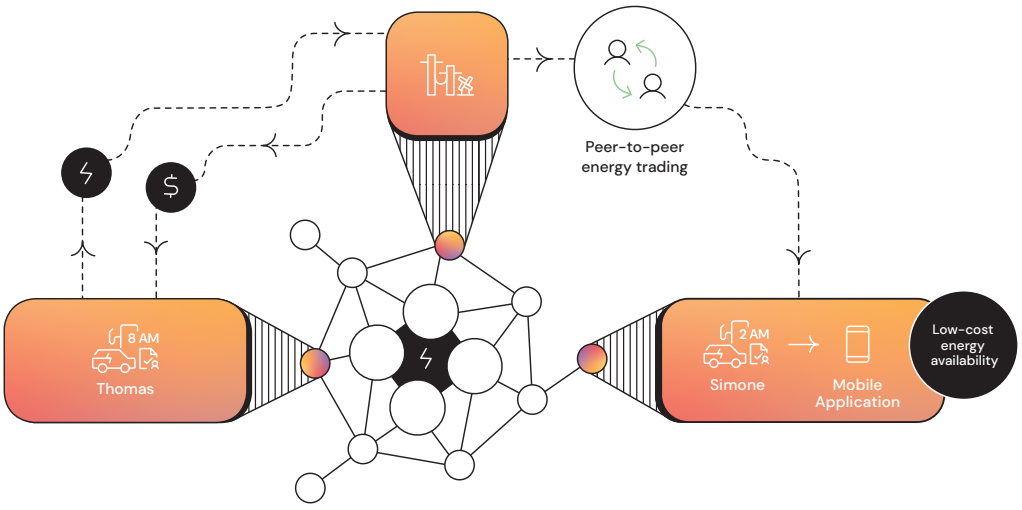


Figure 13. Overview of how vehicle-to-grid systems might interact with a DEG enabled infrastructure.

6.6
Peer-to-peer energy trading: Optimizing local value

In a suburban neighbourhood in a country with a high penetration of rooftop PV, **Maya** (an energy producer with a small wind turbine) and **Sam** (an EV owner) decide to transact energy directly. Maya's turbine often produces surplus power at night, while Sam's EV needs charging at the same time.

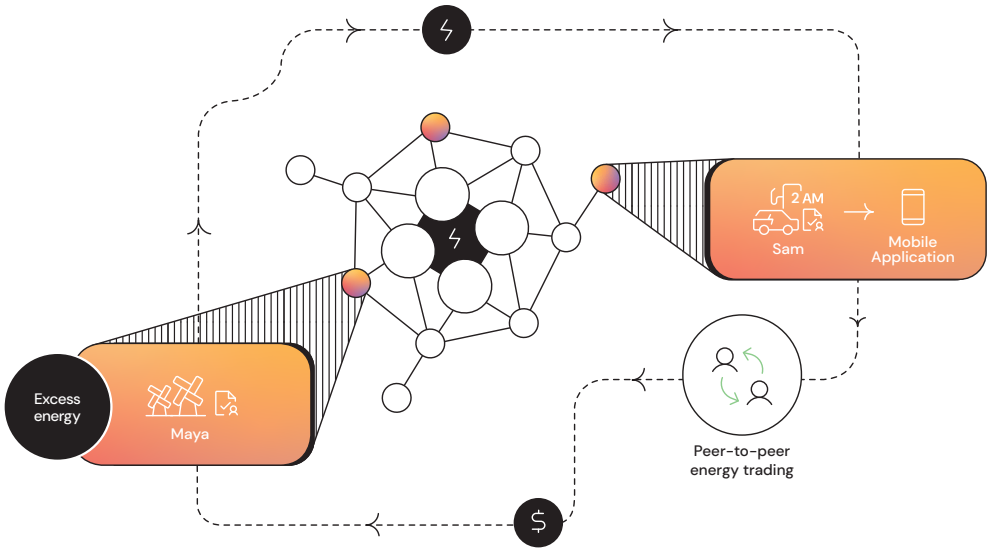
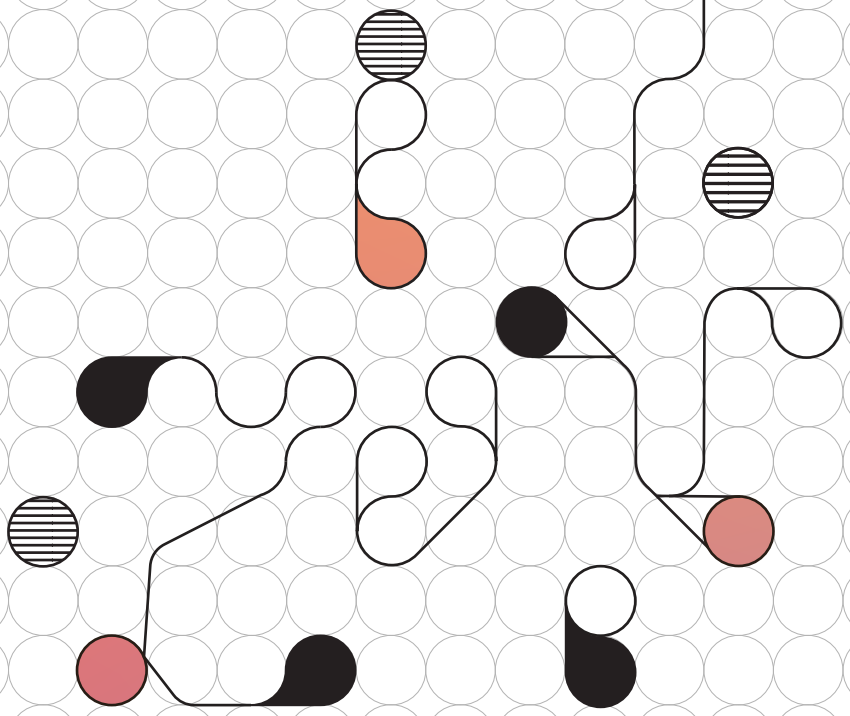


Figure 14. Illustration of a peer-to-peer solar trading environment with verifiable exchange and local balancing.

Without the Digital Energy Grid, Maya might have sold that surplus to the grid at a low feed-in tariff, while Sam would pay regular retail rates to charge. But through a peer-to-peer (P2P) trading platform integrated with DEG, Maya lists her extra energy at a dynamically adjusted, slightly discounted price. Sam's charger checks the marketplace, sees Maya's offer, and purchases her wind power directly. Both sides benefit—Maya earns more than she would from the standard grid rate, Sam saves on charging, and the local grid experiences less congestion. All transactions, from metering to settlement, are recorded with cryptographic certainty, ensuring trust without the overhead of traditional intermediaries.

07

DEG and AI:
A symbiotic relationship



Artificial intelligence (AI) is emerging as a general-purpose technology—comparable in impact to the steam engine or electricity—that is set to transform both the global economy and the energy system. AI advances require cutting-edge computations and vast, high-quality datasets, raising questions about how to meet its increasing energy needs. Simultaneously, AI offers opportunities to reshape the energy sector by enhancing grid flexibility, boosting efficiency, accelerating clean energy adoption, and driving decarbonization efforts. In this intertwined landscape, the Digital Energy Grid (DEG) plays a pivotal role, powering AI while enabling AI to optimize energy systems.

7.1 DEG powers AI

Unifying data

AI models in the energy sector rely heavily on high-quality, accessible data. Digitalization has led to exponential data growth—smart meters generate far more data than analog ones, and wind turbines may produce over 400 billion data points annually. Yet, much of this information is fragmented and hard to access, stalling AI research. DEG addresses this by offering shared data models and standardized frameworks that allow innovators to access high-fidelity datasets under proper permissions. By unifying fragmented data streams into verifiable, searchable, and structured datasets, DEG reduces the costs of inference and training, thereby enhancing the effectiveness of AI tools used for demand prediction and grid optimization. Its layered architecture—spanning transaction, information, and energy flows—facilitates microdata generation and large-scale sharing, ensuring scalable and efficient AI deployments.

Sustainable data centers

Data centers are the backbone of AI computations, yet they consume enormous amounts of energy. As noted by the IEA director, “there is no AI without energy.” Many technology companies are now powering their data centers with renewable energy to meet sustainability goals. However, rapid data center construction challenges the energy sector to anticipate rising electricity demands, locate optimal sites, and integrate these loads into the grid. DEG provides full visibility into grid operations and enables micro and bidirectional energy transactions, offering real-time monitoring of energy flows. This insight helps technology companies strategically select data center locations based on regional energy loads, renewable availability, and grid stability. DEG also streamlines permitting through digital load signatures, programmable policies, and automated monitoring, while regulators benefit from real-time, verifiable data feeds that enforce policies with cryptographic security.

Smart demand response for data centers

Integrated measurability within DEG enables technology companies to determine in real time whether the power received from the grid comes from renewable sources. This precision allows non-urgent tasks to be rescheduled to times when renewable energy is available, matching each energy unit to a "zero-carbon electron." Such carbon-aware computing practices not only lower energy consumption and emissions but also contribute to grid stability by smoothing demand spikes.

Transactional AI and an agentic energy ecosystem

The increasing decentralization and dynamism of energy systems create complexity and coordination challenges. DEG overcomes these by unifying fragmented systems, streamlining operations, and reducing coordination overhead. When paired with AI-powered automation, everyday energy processes become seamlessly optimized in real time. Imagine instructing your home assistant, "Minimize my monthly energy costs," and having it analyze real-time energy prices, demand, and supply data to execute secure, immutable micro-contracts automatically. This is possible because DEG unifies local P2P energy markets, smart appliances, and grid operations.

Cities, too, can benefit from AI agents managing complex systems. For instance, an AI-enabled public transport system integrated with DEG could predict passenger demand, optimize routes, and manage the charging schedules of electric buses. In such a system, an e-bus driver might receive an AI-assisted recommendation to locate and reserve the nearest charging station automatically. Similarly, businesses can use AI agents to automate energy transactions, prioritize green energy suppliers, verify credentials, and negotiate contracts—all while ensuring regulatory compliance.

7.2 AI elevates DEG

Demand shifting and scheduling

DEG's unified data allows AI models to be trained on large datasets, including meteorological forecasts and historical grid usage, to predict energy demand with high precision. A smart grid powered by AI can analyze hourly trends to identify surges in energy usage and automatically schedule energy distribution. Stored surplus energy in batteries, for example, can be deployed during peak demand periods, or surplus energy can be sold back to the grid when supply exceeds demand. This real-time forecasting and scheduling, enabled by DEG's

interoperability across grids, distributed energy resources (DERs), and smart appliances, results in a more resilient and efficient energy market while driving further investment in renewable energy.

Identifying decarbonisation opportunities

AI models, using data from DEG, can combine satellite imagery with information on energy loads and transmission lines to identify optimal locations for solar installations and other renewable projects. By mapping grid networks and analyzing energy transactions, AI helps optimize the mix of renewable energy sources and improve the efficiency of both fossil fuel and renewable power plants. This strategic approach supports low-emission policies and accelerates decarbonization efforts.

Predictive maintenance and anomaly detection

One of AI's most promising applications is predictive maintenance. With DEG data, AI can continuously monitor grid performance, detect anomalies, and predict failures before they cause major disruptions. This proactive approach minimizes downtime and reduces operational costs. Regulators, too, can employ AI-driven anomaly detection to monitor data-sharing frameworks, ensuring compliance with consent protocols and protecting sensitive information.

Seamless user interaction

Advanced voice interfaces and chatbots powered by large language models (LLMs) help non-experts navigate complex energy systems. For example, a community energy manager in a microgrid might simply say, "Balance the load to maximize solar energy usage," and the system, with minimal human supervision, will adjust power distribution autonomously. This integration of natural language processing with DEG's data interoperability makes energy management more accessible and transparent.

7.3 A symbiotic future



Artificial intelligence and the Digital Energy Grid are set to mutually reinforce one another. AI's growing computational capabilities and data needs drive the development of sustainable energy infrastructures, while DEG's robust data-sharing and real-time optimization empower AI to become more adaptive and efficient.

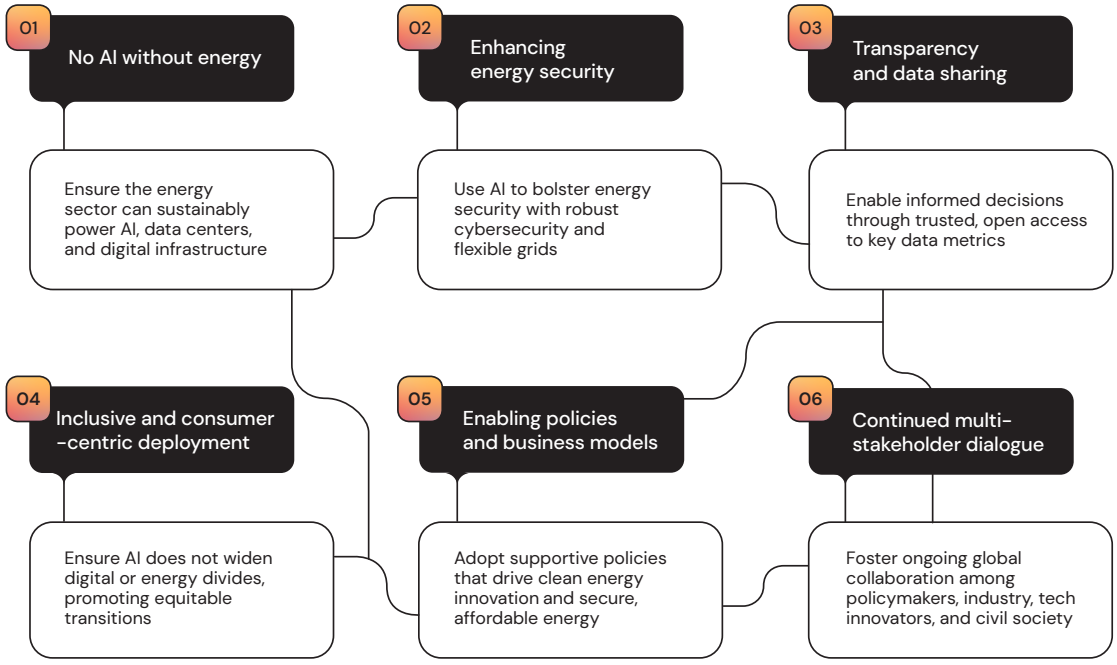
As Fatih Birol noted, “Balancing AI’s energy demands with its transformative potential requires unprecedented cooperation across sectors and borders.”

The future of energy and AI demands a global effort uniting industry leaders, policymakers, and innovators. Through effective integration of AI and energy systems, we can reduce emissions, drive sustainability, and generate significant social and economic value, ensuring a prosperous and greener future for all.

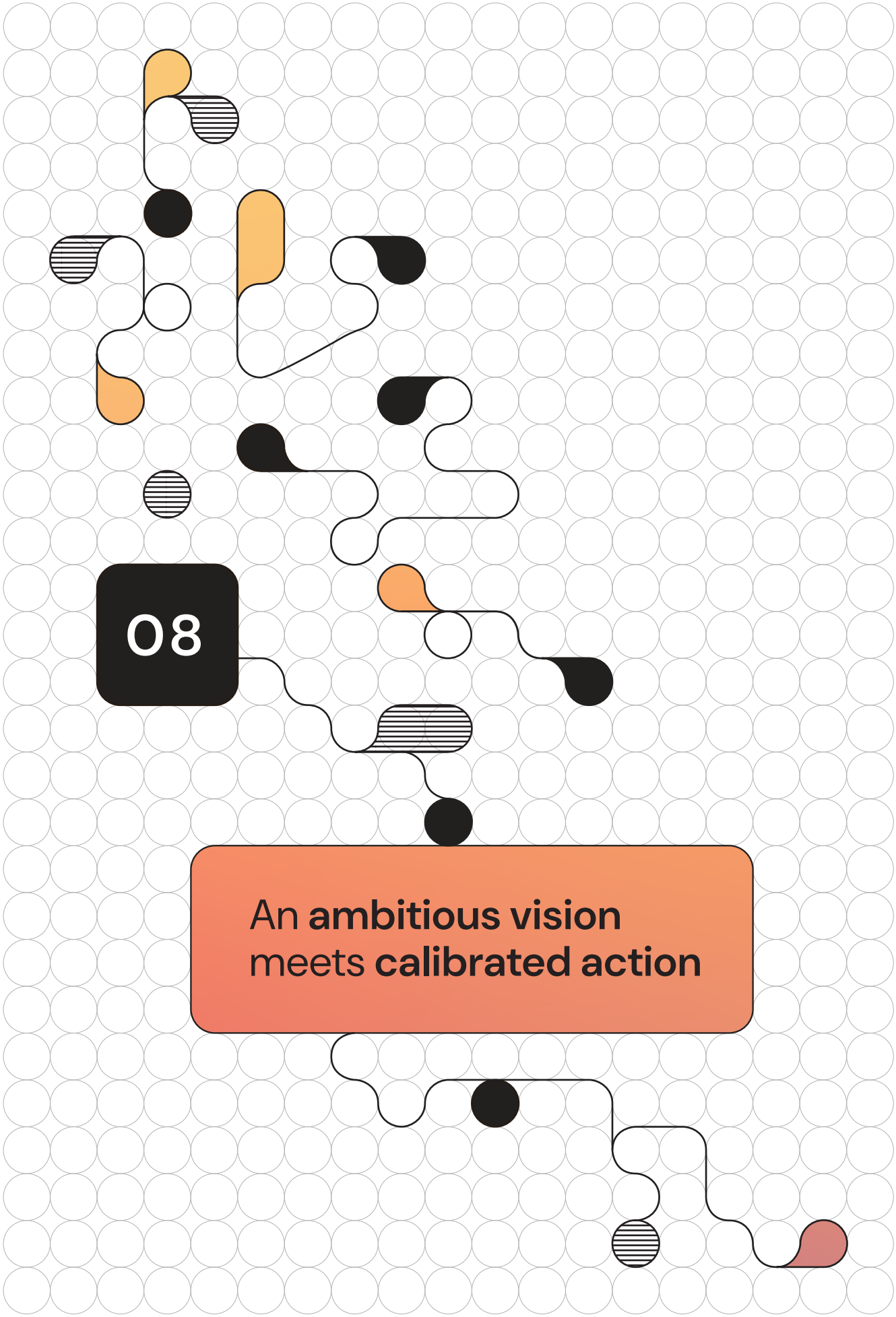
Principles for energy and AI

These principles, as discussed at the IEA Global Conference on Energy and AI[29], offer a sound framework to unlock the potential of AI vis-à-vis the energy sector. A unified energy system that generates useful data, is interoperable, transparent and secure makes it possible for efficient integration of AI-based solutions in the sector.

DEG is one such framework that could help realise some of the desirable outcomes aligned with these principles.



By uniting DEG and AI, these principles are realized, creating energy systems that are efficient, secure, inclusive, and transparent.


08
An ambitious vision
meets calibrated action

DEG represents an ambitious concept, aiming to unify energy systems globally under a common digital infrastructure. Yet its success depends on practical, incremental steps that acknowledge local contexts, variable market readiness, and differing policy environments. Balancing ambition with a pragmatic roadmap allows the framework to adapt, experiment, and evolve in stages without losing sight of the broader objective: a trusted, scalable, and inclusive energy ecosystem.

8.1 DEG vision is context-aware, future-ready, and asynchronous

Every region and market stand at a unique juncture. Some countries already have advanced digital payment systems, widespread broadband, and robust grid automation, while others are still expanding electrification or introducing basic metering infrastructure. DEG's design accommodates this diversity through a modular architecture and asynchronous rollout.

Context-aware adoption

Regions can begin by integrating the most relevant features. For instance, a place already rich in distributed solar might start by assigning digital identities to rooftop installations. A grid with widespread smart meters may prioritize machine-readable, real-time data for demand response. Since DEG is technology-neutral, no single path is forced on every participant.

Future-ready, incremental growth

Implementations can evolve over time. A utility that initially uses DEG for verifying new solar interconnections can expand to real-time market integration or advanced services like peer-to-peer energy trading. Once foundational elements (universal identity, verifiability, machine-readable metadata) are in place, layering advanced capabilities becomes more straightforward.

Asynchronous timelines

Not all markets move in lockstep. Early adopters can pioneer robust governance and pilot programs that later adopters build upon. Interoperable protocols ensure that even a partial rollout can link up with broader networks elsewhere, achieving global reach piece by piece.

A working example of these principles is beginning to take shape through the Unified Energy Interface (UEI)[30], an open network enabled by **Beckn Protocol**[31] for electric vehicle charging, peer-to-peer trading, and demand-response actions. Initiatives like these illustrate how multiple stakeholders across regions can gradually align to a shared vision while tackling real-world challenges.

8.2 DEG as a catalyst for a unified energy future

DEG is neither a single product nor a monolithic technology solution. It is a flexible framework that unifies multiple stakeholders, bridging existing systems and paving the way for new approaches. It complements ongoing advancements in grid infrastructure, hardware design, and energy storage breakthroughs; it does not replace them.

Not a silver bullet

Addressing climate goals, universal energy access, and rapid demand growth also requires major strides in hardware uniformity, grid upgrades, policy reforms, and market-based incentives. Physical generation, distribution, and storage assets must still be built or retrofitted.

Amplifier of innovations

The digital backbone that DEG offers can magnify the impact of emerging technologies. Advanced battery chemistries or new cost-effective solar modules become far more valuable when integrated into a coordinated marketplace, tracked by verifiable data, and governed by transparent rules. Similarly, local pilot programs can scale more easily if they adhere to widely recognized, open standards.

Adjusting to uncertainty

The world's energy landscape is diverse, with unpredictable changes in economics, politics, and technology. DEG's modular design is meant to handle this uncertainty. A wide array of stakeholders—from city governments to rural cooperatives—can adopt the framework in steps, each focusing on their immediate objectives while contributing to a larger, global vision.

8.3
A robust governance framework is crucial for transparency and adoption

A universal digital infrastructure that touches grids, markets, and consumers at scale requires a strong governance model. Public and private innovation can thrive on robust public rails that ensure inclusivity, fairness, and competition. Effective governance is essential to establish guardrails that protect stakeholders' interests without stifling creativity or free market dynamics. By balancing regulation with open access, the system should foster equitable opportunities while safeguarding against exclusion and promoting sustainable growth.

A notable strength of DEG is the ability to embed policies directly into the infrastructure itself. Regulatory requirements—like emissions caps, dynamic tariffs, or renewable portfolio standards—can be encoded as digital rules that operate autonomously. This lowers enforcement costs and ensures that compliance processes scale in tandem with expanding networks. At the same time, DEG enables local regulators and governments to maintain autonomy and sovereignty in implementing the framework within their jurisdictions, ensuring that deployment aligns with regional policies, priorities, and energy transition goals.

Policymakers play a crucial role in shaping resilient digital energy systems by fostering innovation, interoperability, and security. This involves creating adaptable regulatory frameworks, promoting open standards for seamless integration, and aligning public utilities with the vision of unified energy systems. Encouraging consumer participation and mobilising private sector investment are key to driving demand-side flexibility and scaling smart technologies. Additionally, robust cybersecurity measures must be prioritised to protect critical infrastructure and ensure the resilience of these systems in an increasingly interconnected energy landscape.

8.4
From vision to action

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DEG cannot develop in a vacuum. To bring the vision to life, a broad coalition of utilities, technology companies, regulators, financiers, and civil society groups must unite in shaping the underlying infrastructure. Regardless of where one stands in the digital maturity journey, DEG provides a scalable and adaptable framework that integrates existing systems while enabling future advancements. This paper invites all stakeholders—governments, private enterprises, community organizations, and academic institutions—to engage in building and refining the DEG approach. Together, the aim is to overhaul the current system's inefficiencies, harness new opportunities, and move toward a more sustainable and equitable energy future.

Appendix

Annexure: Lessons from Population-Scale Digital Infrastructure

Population-scale digital infrastructure demonstrates how interoperability, universality, user-centricity, and modularity can transform economies and enhance inclusion. These global examples showcase the impact of innovative, scalable systems on financial services, welfare delivery, and economic growth.

India's Unified Payments Interface (UPI)

UPI revolutionised digital payments by enabling micro-level transactions with values as low as one cent. Growing from 0.8 billion transactions in January 2018 to 14.4 billion in July 2024, UPI now handles nearly 80% of India's digital payments[32]. Its success stems from design principles such as interoperability, low cost, and ease of access, demonstrating how scalable infrastructure can drive financial inclusion, innovation and economic participation.

Brazil's Pix payment system

Pix, launched in November 2020, rapidly expanded digital payment adoption in Brazil, adding 71.5 million users by December 2022 and serving 153 million individuals (around 75% of the population) and 15 million businesses by March 2024[33]. By focusing on speed, flexibility, and interface standardisation, Pix enabled small merchants and startups to thrive, highlighting how infrastructure thinking can bridge financial and technological divides.

Estonia's X-Road: A pioneer in digital governance

Estonia's X-Road, launched in 2001, exemplifies transformative power at scale. It has digitised 99% of government services, saving 1,407 years of working time each year[34]. This comprehensive digital infrastructure has far-reaching effects: attracting global entrepreneurs through e-residency, inspiring similar systems in other countries, and positioning Estonia as a leader in digital governance. The X-Road system has enabled rapid response during crises and contributed significantly to Estonia's reputation as an innovative digital society.

Open networks

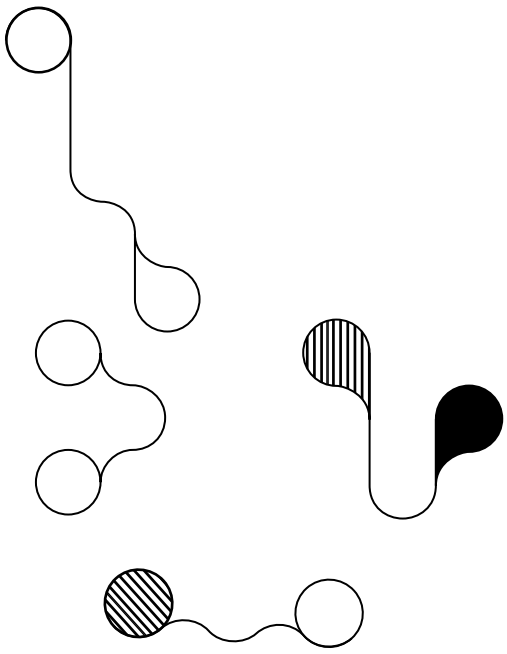
The implementation of Open Transaction Networks (OTNs)[35] for commerce, mobility and other sectors particularly through the Open Network for Digital Commerce (ONDC), exemplifies the lowering of transaction costs and barriers to entry, thereby cultivating an environment ripe for innovation, competition and market expansion.

The ONDC (ondc.org)[36], underpinned by Beckn protocol (becknprotocol.io), is pioneering a significant shift in the transaction economy, demonstrating how open, protocol-based systems can fundamentally alter market dynamics and foster inclusive growth in domestic and global trade.

Direct benefit transfers

Welfare programs leveraging digital infrastructure, such as India's PAHAL[37] for LPG subsidies, showcase how direct benefit transfers reduce inefficiencies, optimise targeting, and enhance transparency. These systems provide valuable models for subsidy reforms in other sectors, such as energy.

Together, these initiatives highlight the profound potential of population-scale digital infrastructure to drive innovation, improve efficiency, and foster inclusion across global economies. These examples offer policymakers actionable insights for designing transformative systems in diverse contexts.



Glossary

Battery Energy Storage System (BESS)	Technology that stores energy for later use, aiding in grid stability and the integration of renewable energy sources.
Carbon Credits	Tradable permits that allow the holder to emit a certain amount of carbon dioxide, promoting emissions reduction.
Cryptographic Verifiability	The use of cryptographic techniques to ensure data integrity, authenticity, and tamper-proof records.
Decentralisation	The distribution of energy generation and management across various smaller systems or entities instead of a central authority.
Demand Response (DR)	Programs enabling consumers to adjust their electricity usage during peak demand periods in response to market signals or incentives.
Digital Energy Grid (DEG)	A vision for unified, scalable digital infrastructure that integrates physical energy flows with digital information and transaction flows across the energy ecosystem.
Dynamic Load Balancing	Adjusting energy supply and demand in real-time to maintain grid stability and efficiency.
Dynamic Pricing	Electricity pricing that fluctuates based on supply and demand, encouraging efficient energy usage and integration of renewables.
Energy-as-a-Service (EaaS)	Business models where energy-related services (e.g., storage, optimization) are offered to customers without requiring ownership.
Energy Management System (EMS)	Systems that monitor, control, and optimize energy use and generation.
Energy Market Operator	Entities that manage the trading, supply, and demand of energy within a grid to ensure stability and efficiency.
Energy Sovereignty	The ability of a nation, community, or individual to control and manage its energy resources sustainably and independently.
Energy Transition	The shift from fossil fuel-based energy systems to renewable and low-carbon energy sources to achieve sustainability goals.
Grid Congestion	Bottlenecks in the power system due to limited transmission capacity, leading to inefficiencies and increased costs.

Grid Orchestration Software	Tools that facilitate the management of distributed energy resources and grid operations in real-time.
Interoperability	The ability of different systems, devices, and organizations to work together seamlessly, using common protocols and standards.
Layer 1 and Layer 2 Standards	Layer 1: Purpose-specific standards for data formats or systems (e.g., JPEG for images). Layer 2: General-purpose frameworks that unify and connect Layer 1 standards (e.g., HTML for web content).
Machine Readable Data	Data formatted in a way that can be automatically processed by machines without human intervention, enabling automation and analytics.
Microgrid	A localized group of energy resources and loads that can operate independently or in connection with the larger grid.
Negative Pricing	Situations where electricity producers must pay to offload excess generation due to lack of demand or grid constraints.
Peer-to-Peer (P2P) Energy Trading	A decentralized model where individuals or entities trade energy directly without intermediaries.
Programmability	The capacity to configure, control, and automate energy transactions and operations digitally.
Smart Meters	Digital devices that measure and report electricity usage in real-time, enabling better energy management and billing.
Tokenised Carbon Credits	Carbon credits represented as digital tokens, allowing for efficient trading and verification.
Universal Identity	A unique, verifiable identifier assigned to energy assets, actors, and systems to ensure interoperability and traceability.
Vehicle-to-Grid (V2G)	Technology that allows energy to be returned from electric vehicle batteries to the grid during peak demand.
GW (Gigawatt)	A unit of power equal to one billion watts, commonly used to measure the output of large power plants or the capacity of electricity grids.
TWh (Terawatt-hour)	A unit of energy equal to one trillion watt-hours, often used to describe the annual electricity consumption of countries or large systems.

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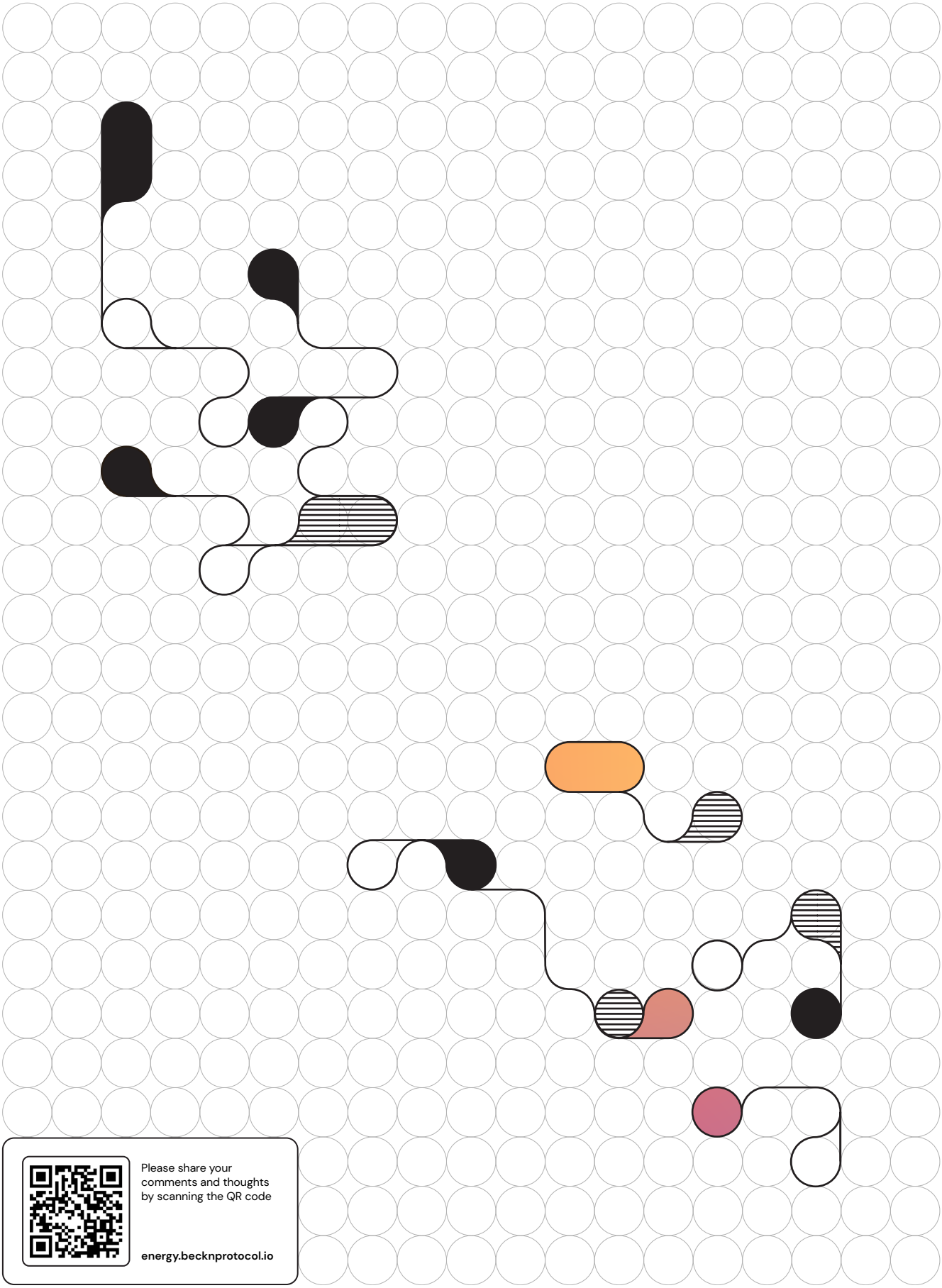
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